

ICS 01.040.91; 91.100.23

DMS 1728:2021

First edition

ISO 13006:2018

IDT

DRAFT MALAWI STANDARD

**Ceramic tiles — Definitions,
classifications, characteristics and
marking**

NOTE: This is a Draft Malawi Standard and should not be regarded or used as a Malawi Standard

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Ceramic tiles — Definitions, classification, characteristics and marking

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DRAFT MALAWI STANDARD

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FOREWORD

This Draft Malawi Standard is an adoption of ISO 13006:2018, *Ceramic tiles — Definitions, classification, characteristics and marking*.

Acknowledgement is made for the use of the above standard.

TECHNICAL COMMITTEE

This Draft Malawi Standard was developed by the Technical Committee MBS/TC 3, *Masonry and natural stones including bricks, tiles and stones*, and the following companies and organisations were consulted:

Blantyre City Council;
Built Environs Limited;
City Building Contractors;
Dedza Pottery;
Fargo Limited;

Focus Tiles Trading;
Gimtec Construction Limited;
LafargeHolcim Malawi;
Lilongwe City Council;
Malawi Bureau of Standards
Malawi Builders and Civil Engineering Contractors and Allied Traders Association;
Malawi Housing Corporation;
Malawi Institute of Engineers;
Ministry of Lands, Housing and Urban development;
Malawi University of Business and Applied Sciences;
Ministry of Transport and Public Works;
Malawi University of Science and Technology;
National Construction Industry Council;
O.G. Construction Limited; and
Terrastone Construction Limited.

NOTICE

The approved Malawi Standard shall be revised every five years or earlier when necessary in order to keep abreast of progress. Comments are welcome and shall be considered when the approved standard is being revised.

DRAFT MALAWI STANDARD

DRAFT MALAWI STANDARD**Ceramic tiles – Definitions, classification, characteristics and marking**

1 SCOPE

This Draft Malawi Standard defines terms and establishes classifications, characteristics and marking requirements for ceramic tiles of the best commercial quality (first quality). This document is not applicable to tiles made by other than normal processes of extrusion or dry pressing. It is not applicable to decorative accessories or trim such as edges, corners, skirting, capping, coves, beads, steps, curved tiles and other accessory pieces or mosaics (i.e., any piece that can fit into a square, the side of which is less than 7 cm).

NOTE – DMS 1701 (all parts) describes the test procedures required to determine the product characteristics listed in this document. **DMS 1701** is a multi-part standard, each part describes a specific test procedure or related matter.

2 NORMATIVE REFERENCES

The following standards contain provisions which, through reference in this text, constitute provisions of this Draft Malawi Standard. All standards are subject to revision and, since any reference to a standard is deemed to be a reference to the latest edition of that standard, parties to agreements based on Draft Malawi Standard are encouraged to take steps to ensure the use of the most recent editions of the standards indicated below. Information on currently valid national and international standards can be obtained from the Malawi Bureau of Standards.

DMS 1701-1, *Ceramic tiles — Part 1: Sampling and basis for acceptance*;

DMS 1701-2, *Ceramic tiles — Part 2: Determination of dimensions and surface quality*;

DMS 1701-3, *Ceramic tiles — Part 3: Determination of water absorption, apparent porosity, apparent relative density and bulk density*;

DMS 1701-4, *Ceramic tiles — Part 4: Determination of modulus of rupture and breaking strength*;

DMS 1701-5, *Ceramic tiles — Part 5: Determination of impact resistance by measurement of coefficient of restitution*;

DMS 1701-6, *Ceramic tiles — Part 6: Determination of resistance to deep abrasion for unglazed tiles*;

DMS 1701-7, *Ceramic tiles — Part 7: Determination of resistance to surface abrasion for glazed tiles*;

DMS 1701-8, *Ceramic tiles — Part 8: Determination of linear thermal expansion*;

DMS 1701-9, *Ceramic tiles — Part 9: Determination of resistance to thermal shock*;

DMS 1701-10, *Ceramic tiles — Part 10: Determination of moisture expansion*;

DMS 1701-11, *Ceramic tiles — Part 11: Determination of crazing resistance for glazed tiles*;

DMS 1701-12, *Ceramic tiles — Part 12: Determination of frost resistance*;

DMS 1701-13, *Ceramic tiles — Part 13: Determination of chemical resistance*;

DMS 1701-14, *Ceramic tiles — Part 14: Determination of resistance to stains*;

DMS 1701-15, *Ceramic tiles — Part 15: Determination of lead and cadmium given off by glazed tiles*;

DMS 1701-16, *Ceramic tiles — Part 16: Determination of small colour differences*; and

ISO 1006, *Building construction — Modular coordination — Basic module*.

3 TERMS AND DEFINITIONS

For the purposes of this document, the terms and definitions given in **ISO 1006** and the following apply.

3.1

ceramic tile

thin slab made from clays and/or other inorganic raw materials, generally used as covering for floors and walls, usually shaped by extruding (A) or pressing (B) at room temperature, but may be formed by other processes (C), then dried and subsequently fired at temperatures sufficient to develop the required properties

Note 1 to entry: Tiles may be glazed (GL) or unglazed (UGL); they are incombustible and are not affected by light.

3.2

porcelain tile

fully vitrified tile with water absorption coefficient less than or equal to a mass fraction of 0.5 %, belonging to groups Ala and Bla

3.3

glaze

vitrified covering that is impermeable

3.4

engobed surface

clay-based covering with a matt finish which may be permeable or impermeable

Note 1 to entry: A tile with an engobed surface is regarded as an unglazed tile.

3.5

polished surface

surface of glazed and unglazed tile which has been given a glossy finish by mechanical polishing as the last stage of manufacture

3.6

extruded tile

tile, whose body is shaped in the plastic state in an extruder, the column obtained being cut into tiles of pre-determined dimension, and which is designated as Group A

Note 1 to entry: This document classifies extruded tiles as “precision” or “natural”. The classification is dependent upon the different technical characteristics as listed in the individual product standards.

Note 2 to entry: Traditional terms used for extruded products are “split tiles” and “quarry tiles”. They commonly indicate double-extruded and single-extruded tiles, respectively. The term “quarry tiles” only refers to extruded tiles with a water absorption coefficient of a mass fraction not exceeding 6 %.

3.7

dry-pressed tile

tile, formed from a finely milled body mixture and shaped (e.g. in dies or moulds) at high pressure, and which is designated as Group B

3.8

water absorption

E_v

percentage of water impregnating a tile

Note 1 to entry: This is measured in accordance with **DMS 1701-3**.

Note 2 to entry: Water absorption is expressed as a mass fraction of dry mass.

3.9 Sizes

Note 1 to entry: The sizes are only defined for rectangular tiles. If the sizes of non-rectangular tiles are required, they are defined by the smallest rectangle into which they fit.

3.9.1

nominal size

size used to describe the product

Note 1 to entry: See **Figures 1** and **2**.

3.9.2

work size

size of a tile specified for manufacturing to which the actual size shall conform within specified permissible deviations

Note 1 to entry: See **Figures 1** and **2**.

3.9.3

actual size

size obtained by measuring the face of a tile

Note 1 to entry: This is measured in accordance with **DMS 1701-2**.

Note 2 to entry: See **Figures 1** and **2**.

3.9.4

coordinating size

work size plus the joint width

Note 1 to entry: See **Figures 1** and **2**.

3.9.5

modular size

tile and size based on module M, 2 M, 3 M and 5 M and also their multiples or subdivisions, except for tiles with a surface area of less than 9000 mm²

Note 1 to entry: See **ISO 1006**, where 1 M = 100 mm.

Note 2 to entry: See **Figures 1** and **2**.

3.9.6

non-modular size

size not based on module M

Note 1 to entry: See **ISO 1006**, where 1 M = 100 mm.

Note 2 to entry: Tiles of these sizes are those commonly used in most countries.

Note 3 to entry: See **Figures 1** and **2**.

3.9.7

tolerance

difference between the permissible limits of size

Note 1 to entry: See **Figures 1** and **2**.

3.10

spacer lug

projection that is located along certain edges of tiles so that where two tiles are placed together, in line, the lugs on adjacent edges separate the tiles by a distance of not less than the specified width of the joint

Note 1 to entry: See **Figure 2**.

3.11

rectified tile

ceramic tile that, after firing, is subjected to a precise mechanical finishing of the edges

3.12

back feet

parallel ridges running across the back surface of some exterior wall tiles which possess a geometry intended to facilitate an interlocking connection between tile and cement mortar

Note 1 to entry: See Figure 3.

Note 2 to entry: Back feet dimensions shall be measured with vernier calipers, and each specimen in a 10-tile sample is required to satisfy the requirements in **Annexes A to H** and **Annexes J to M**.

4 CLASSIFICATION

4.1 Basis of classification

Ceramic tiles are divided into groups according to their method of manufacture and their water absorption (see **3.8** and **Table 1**). The groups do not presuppose the usage of the products.

4.2 Methods of manufacture

The following are the two methods of manufacture:

- method A, extruded tiles (see **3.6**);
- method B, dry-pressed tiles (see **3.7**).

4.3 Water absorption according to group

4.3.1 General

The following are the three groups according to water absorption, E_v .

4.3.2 Subdivision of the three groups

The three groups are divided into tiles of low, medium and high water absorption, namely Groups I, II and III, respectively.

- a) Tiles of low water absorption, i.e. absorption coefficient less than or equal to a mass fraction of 3 %, $E_v \leq 3$ %, belong to Group I. Group I consists of the following:
 - 1) for extruded tiles
 - i) $E_v \leq 0.5$ % (Group AIa), and
 - ii) 0.5 % $< E_v \leq 3$ % (Group AIb);
 - 2) for dry-pressed tiles:
 - i) $E_v \leq 0.5$ % (Group BIa);
 - ii) 0.5 % $< E_v \leq 3$ % (Group BIb).
- b) Tiles of medium water absorption, i.e. 3 % $< E_v \leq 10$ %, belong to Group II. Group II consists of the following:
 - 1) for extruded tiles'
 - i) 3 % $< E_v \leq 6$ % [Group AIIa, Subgroups (Parts) 1 and 2; see **Annex B** for Subgroup (Part) 1 or **Annex C** for Subgroup (Part) 2], and
 - ii) 6 % $< E_v \leq 10$ % [Group AIIb, Subgroups (Parts) 1 and 2; see **Annex D** for Subgroup (Part) 1 or **Annex E** for Subgroup (Part) 2];
 - 2) for dry-pressed tiles
 - i) 3 % $< E_v \leq 6$ % (Group BIIa), and

ii) $6\% < E_v \leq 10\%$ (Group BIIb).

c) Tiles of high water absorption, i.e. $E_v > 10\%$, belong to Group III.

5 CHARACTERISTICS

The characteristics for different applications of ceramic tiles are given in **Table 2**.

6 SAMPLING AND BASIS FOR ACCEPTANCE

The sampling and basis for acceptance shall be in accordance with that presented in **DMS 1701-1**.

7 REQUIREMENTS

Dimensional and surface quality requirements and physical and chemical properties shall be as given in the relevant/specific annex, of **Annexes A to H** and **Annexes J to M**, for each tile class.

NOTE – Annexes I, O and Q are intentionally left blank. This is as a convenience to producers in order to avoid changes to packaging and related costs. In addition, at the time of publication, the market offers consumers various products identified by the matching annex headings of this document.

8 MARKING AND SPECIFICATIONS

8.1 Marking

Tiles and/or their packaging are required to bear the following marking:

- a) manufacturer's mark and/or trademark and the country where the tile was manufactured (i.e. country of origin, as determined by the relevant international regulation);
- b) indication of first quality;
- c) type of tile and reference to the appropriate group and/or annex of **Annexes A to H** and **Annexes J to M** covering the specific group of tile;
- d) nominal and work sizes, and "M" if modular;
- e) nature of the surface, i.e. glazed (GL) or unglazed (UGL);
- f) any surface treatment applied after firing;
- g) total number of tiles in the package;
- h) manufacturer's production run or batch number;
- i) an indication of colour consistency, as defined by the manufacturer;
- j) total dry weight which the tiles and their packaging shall not exceed.

EXAMPLE 1

The ABC Tile Co., Made in China, Ceramic Tile of First Quality Extruded tile, DMS 1728:2021, Annex A, Group Alb, Precision

M, 25 cm × 12.5 cm (S_w 240 mm × 115 mm × 12.5 mm), UGL, Protective Sealer Added
20 pcs/box, batch A50, 0100-Gray, Max dry weight: 20 kg

EXAMPLE 2

The 123 Tile Co., Made in Spain, Ceramic Tile of First Quality

Dry-pressed porcelain tile, DMS 1728:2021, Annex G, Group BIa, Rectified 60 cm × 60 cm (S_w 598 mm × 598 mm × 10 mm) UGL

3 pcs/box, batch B52, 0590-Slate, Max dry weight: 16 kg

EXAMPLE 3

The XYX Tile Co., Made in Indonesia, Ceramic Tile of First Quality Dry-pressed tile, DMS 1728:2021, Annex H, Group BIb Non-rectified 30 cm × 30 cm (Sw 299 mm × 299 mm × 10 mm) GL
10 pcs/box, batch C60, 0320-Ivory, Max dry weight: 20 kg

Each tile conforming to this document is required to bear on its reverse side or edge, the country where it was manufactured.

8.2 Product literature

Product literature for tiles intended for use on floors shall state the abrasion class or the place of use of glazed tiles.

8.3 Specifications

Tiles shall be designated by the following:

- a) the method of shaping;
- b) the relevant group and/or annex of **Annexes A to H** and **Annexes J to M** covering the specific group of tile
- c) nominal and work sizes, and “M” if modular;
- d) the nature of the surface, i.e. glazed (GL) or unglazed (UGL);
- e) the addition of back feet, if required.

EXAMPLE 1 Precision extruded tile, DMS 1728:2021, Annex M, AIa M 25 cm × 12.5 cm (Sw 240 mm × 115 mm × 10 mm) GL.

EXAMPLE 2 Natural extruded tile, DMS 1728:2021, Annex A, Aib 15 cm × 15 cm (Sw 150 mm × 150 mm × 12.5 mm) UGL.

EXAMPLE 3 Dry-pressed tile, DMS 1728:2021, Annex G, BIa M 25 cm × 12.5 cm (Sw 240 mm × 115 mm × 10 mm) GL.

EXAMPLE 4 Dry-pressed tile, DMS 1728:2021, Annex L, BIII 15 cm × 15 cm (Sw 150 mm × 150 mm × 12.5 mm) UGL.

9 ORDERING

Each time that an order is placed, items, such as size, thickness, type of surface, colour, profile, abrasion class for glazed tiles and other properties, shall be agreed upon by the parties concerned.

Table 1 — Classification of ceramic tiles with respect to water absorption and shaping

1	2	3	4	5
Shaping	Group I $E_v \leq 3\%$	Group IIa $3\% < E_v \leq 6\%$	Group IIb $6\% < E_v \leq 10\%$	Group III $E_v > 10\%$
A Extruded	Group AIa $E_v \leq 0.5\%$ (see Annex M)	Group AIIa-1 ^a (see Annex B)	Group AIIb-1 ^a (see Annex D)	Group AIII (see Annex F)
	Group AIb $0.5\% < E_v \leq 3\%$ (see Annex A)	Group AIIa-2 ^a (see Annex C)	Group AIIb-2 ^a (see Annex E)	
B Dry pressed	Group BIa $E_v \leq 0.5\%$ (see Annex G)	Group BIIa (see Annex J)	Group BIIb (see Annex K)	Group BIIIb (see Annex L)
	Group BIb $0.5\% < E_v \leq 3\%$ (see Annex H)			

a Groups AIIa and AIIb are divided into two subgroups (Parts 1 and 2) with different product specifications.
b Group BIII covers glazed tiles only. There is a low quantity of dry-pressed unglazed tiles produced with water absorption greater than 10 % mass fraction, which is not covered by this product group.

Table 2 — Characteristics required for different applications

1	2	3	4	5	6
Characteristics	Floor		Wall		Test
Dimensions and surface quality	Interior	Exterior	Interior	Exterior	Reference
Length and width	X	X	X	X	DMS 1701-2
Thickness	X	X	X	X	DMS 1701-2
Straightness of sides	X	X	X	X	DMS 1701-2
Rectangularity	X	X	X	X	DMS 1701-2
Surface flatness (curvature and warpage)	X	X	X	X	DMS 1701-2
Surface quality	X	X	X	X	DMS 1701-2
Back feet (if specified) ^a				X	Figure 3
Physical property	Interior	Exterior	Interior	Exterior	Reference
Water absorption	X	X	X	X	DMS 1701-3
Breaking strength	X	X	X	X	DMS 1701-4
Modulus of rupture	X	X	X	X	DMS 1701-4
Resistance to deep abrasion — unglazed tiles	X	X			DMS 1701-6
Resistance to surface abrasion — glazed tiles	X	X			DMS 1701-7
Linear thermal expansion ^b	X	X	X	X	DMS 1701-8
Resistance to thermal shock ^b	X	X	X	X	DMS 1701-9
Resistance to crazing — glazed tiles	X	X	X	X	DMS 1701-11
Frost resistance ^c		X		X	DMS 1701-12
Moisture expansion ^b	X	X	X	X	DMS 1701-10
Small colour differences ^b	X	X	X	X	DMS 1701-16

Table 2 (continued)

1	2	3	4	5	6
Characteristics	Floor		Wall		Test
Dimensions and surface quality	Interior	Exterior	Interior	Exterior	Reference
Impact resistance ^b	X	X			DMS 1701-5
Chemical property	Interior	Exterior	Interior	Exterior	Reference
Resistance to staining					DMS 1701-14
— glazed tiles	X	X	X	X	DMS 1701-14
— unglazed tiles ^b	X	X	X	X	DMS 1701-14
Resistance to low concentrations of acids and alkalis	X	X	X	X	DMS 1701-13
Resistance to high concentrations of acids and alkalis ^b	X	X	X	X	DMS 1701-13
Resistance to household cleaners and swim- ming pool salts	X	X	X	X	DMS 1701-13
Lead and cadmium release — glazed tiles ^b	X	X	X	X	DMS 1701-15
a For application to exterior tiles installed by cement mortar, including tunnels, where back feet are specified. b Test method is available. c For tiles intended to be used in situations where frost conditions apply.					

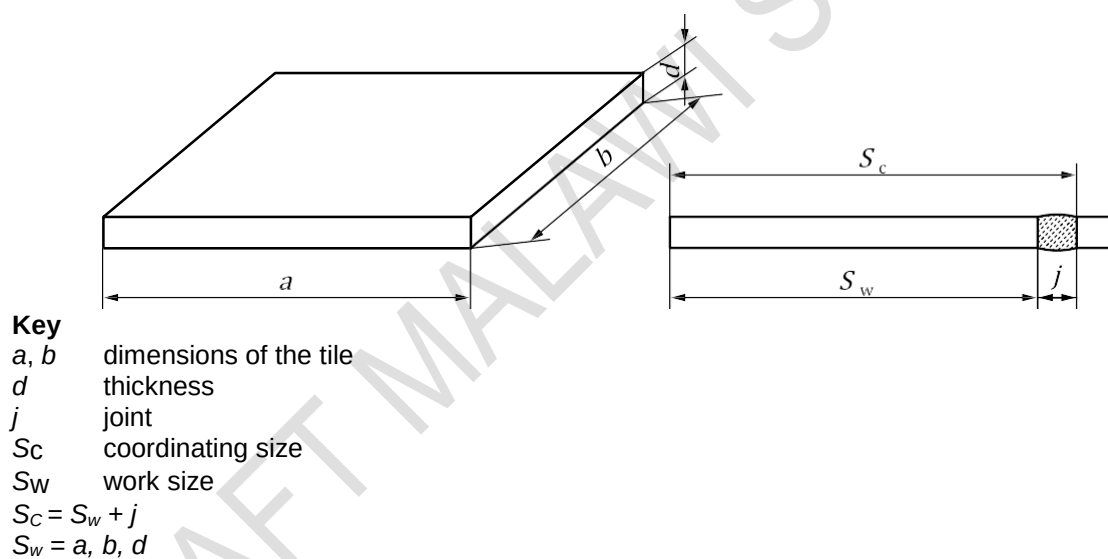
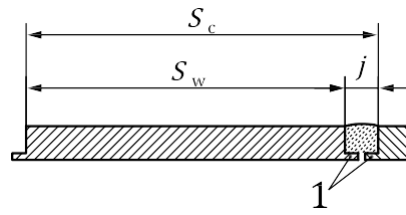


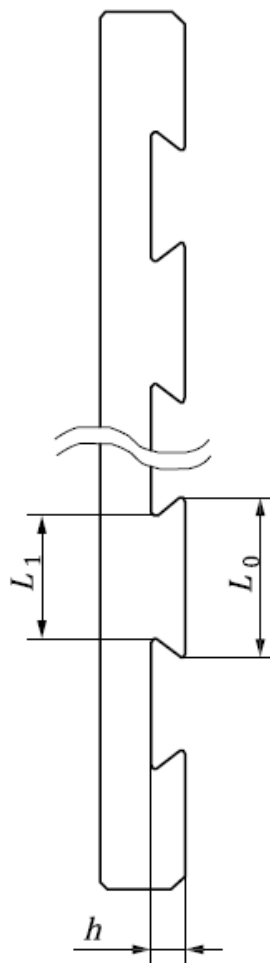
Figure 1 — Tile



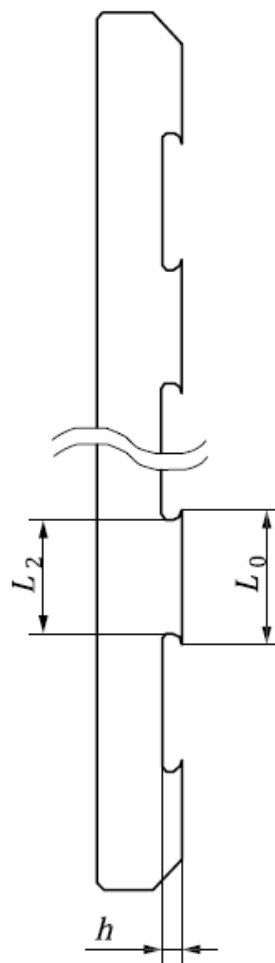
Key

- 1 spacer lugs
- j joint
- S_c coordinating size
- S_w work size
- $S_c = S_w + j$
- $S_w = a, b, d$

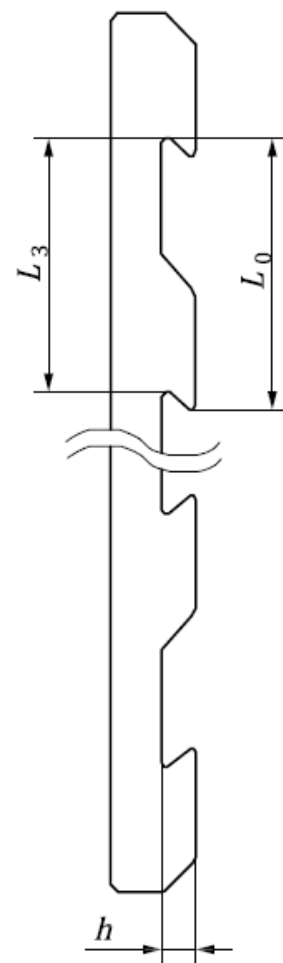
Figure 2 — Tile with spacer lug



a) Example 1



b) Example 2



c) Example 3

Key

- h height
- L length

Figure 3 — Back feet — Examples

Annex A

(normative)

Extruded ceramic tiles with low water absorption $0.5 < E_V \leq 3 \%$ Group AII_b

Dimensional and surface quality requirements and physical and chemical properties are required to be in accordance with **Table A.1**.

Table A.1 — Requirements for extruded ceramic tiles — Group AII_b, $0.5 < E_V \leq 3 \%$

1	2	3	4
Dimensions and surface quality	Precision	Natural	Test
Length and width			
The manufacturer shall choose the work size as follows:			
a) for modular tiles in order to allow a nominal joint width of between 3 mm and 11 mm ^a ;			
b) for non-modular tiles so that the difference between the work size and the nominal size is not more than ± 3 mm.			
The deviation, in percent, of the average size for each tile (two or four sides) from the work size, S_W	$\pm 1.0 \%$ to a maximum of ± 2 mm	$\pm 2.0 \%$ to a maximum of ± 4 mm	DMS 1701-2
The deviation, in percent, of the average size for each tile (two or four sides) from the average size of the 10 test specimens (20 or 40 sides)	$\pm 1.0 \%$	$\pm 1.5 \%$	DMS 1701-2
Thickness			
a) The thickness shall be specified by the manufacturer ⁹			
b) The deviation, in percent, of the average thickness of each tile from the work size thickness ⁹	$\pm 10 \%$	$\pm 10 \%$	DMS 1701-2
Straightness of sides^b (facial sides)			
The maximum deviation from straightness, in percent, related to the corresponding work sizes	$\pm 0.5 \%$	$\pm 0.6 \%$	DMS 1701-2
Rectangularity^b			
The maximum deviation from rectangularity, in percent, related to the corresponding work sizes	$\pm 1.0 \%$	$\pm 1.0 \%$	DMS 1701-2
Surface flatness			
The maximum deviation from flatness, in percent:			
a) centre curvature, related to diagonal calculated from the work sizes;	$\pm 0.5 \%$	$\pm 1.5 \%$	DMS 1701-2
b) edge curvature, related to the corresponding work sizes;	$\pm 0.5 \%$	$\pm 1.5 \%$	DMS 1701-2

Table A.1 (continued)

1	2	3	4
Dimensions and surface quality	Precision	Natural	Test
c) warpage, related to the diagonal calculated from the work sizes.	$\pm 0.8 \%$	$\pm 1.5 \%$	DMS 1701-2
Back feet (if specified)			
a) Height, h , for tiles of surface area, A			
$49 \text{ cm}^2 \leq A < 60 \text{ cm}^2$	Minimum $h = 0.7 \text{ mm}$; Maximum $h = 3.5 \text{ mm}$		Figure 3
$A \geq 60 \text{ cm}^2$	Minimum $h = 1.5 \text{ mm}$; Maximum $h = 3.5 \text{ mm}$		Figure 3
b) Shape	Back feet as specified by the manufacturer and as shown in one of the examples in Figure 3		Figure 3
Example 1 (see Figure 3)	$L_0 - L_1 > 0$		Figure 3
Example 2 (see Figure 3)	$L_0 - L_2 > 0$		Figure 3
Example 3 (see Figure 3)	$L_0 - L_3 > 0$		Figure 3
Surface quality^c	A minimum of 95 % of the tiles are to be free from visible defects which can impair the appearance of a major area of tiles		DMS 1701-2
Physical property	Precision	Natural	Test
Water absorption Percent mass fraction	$0.5 < E_v \leq 3.0 \%$ Individual maximum 3.3 %	$0.5 < E_v \leq 3.0 \%$ Individual maximum 3.3 %	DMS 1701-3
Breaking strength, in Newtons			
a) Thickness $\geq 7.5 \text{ mm}$	Not less than 1 100	Not less than 1 100	DMS 1701-4
b) Thickness $< 7.5 \text{ mm}$	Not less than 600	Not less than 600	DMS 1701-4
Modulus of rupture, in Newtons per square millimetre Not applicable to tiles with breaking strength $\geq 3\,000 \text{ N}$	Minimum 23 Individual minimum 18	Minimum 23 Individual minimum 18	DMS 1701-4
Abrasion resistance			
a) Resistance to deep abrasion of unglazed tiles: removed volume, in cubic millimetres	Maximum 275	Maximum 275	DMS 1701-6
b) Resistance to surface abrasion of glazed tiles intended for use on floors ^d	Report abrasion class and cycles passed	Report abrasion class and cycles passed	DMS 1701-7
Coefficient of linear thermal expansion^e			
From ambient temperature to $100 \text{ }^\circ\text{C}$	Test method available	Test method available	DMS 1701-8
Thermal shock resistance^e	Test method available	Test method available	DMS 1701-9
Crazing resistance: glazed tiles ^f	Required	Required	DMS 1701-11
Frost resistance	Required	Required	DMS 1701-12
Moisture expansion, in millimetres per metre^e	Test method available	Test method available	DMS 1701-10

Small colour differences^e	Plain coloured tiles only where required GL: $\Delta E < 0.75$ UGL: $\Delta E < 1.0$	Plain coloured tiles only where required GL: $\Delta E < 0.75$ UGL: $\Delta E < 1.0$	DMS 1701-16
Impact resistance^e	Test method available	Test method available	DMS 1701-5
Chemical property	Precision	Natural	Test
Resistance to staining			
a) Glazed tiles	Minimum class 3	Minimum class 3	DMS 1701-14
b) Unglazed tiles ^e	Test method available	Test method available	DMS 1701-14
Dimensions and surface quality	Precision	Natural	Test
Resistance to chemicals			
Resistance to low concentrations of acids and alkalis	Manufacturer is to state classification	Manufacturer is to state classification	DMS 1701-13
Resistance to high concentrations of acids and alkalis ^e	Test method available	Test method available	DMS 1701-13
Resistance to household chemicals and swimming pool salts	Minimum B	Minimum B	DMS 1701-13
Lead and cadmium release^e	Test method available	Test method available	DMS 1701-15
a Similar joint widths shall be used to apply to traditional systems based on non-metric sizes. b Not applicable to tiles having curved shapes. c As a result of firing, slight variations from this standard colour are unavoidable. This does not apply to intentional irregularities of colour variation of the face of tiles (which can be unglazed, glazed or partly glazed) or to the colour variation over a tile area which is characteristic for this type of tile and desirable. Spots or coloured dots which are introduced for decorative purposes are not considered a defect. d Use Annex N for the abrasion resistance classification for all glazed tiles intended for use on floors. e Annex P gives information regarding requirements which are non-compulsory but which are listed as “test method available”. f Certain decorative effects can have a tendency to craze. These are to be identified by the manufacturer, in which case, the crazing test given in DMS 1701-11 is not applicable. g Where applicable, tile thickness shall include the addition of back feet heights, as measured in accordance with Figure 3.			

Annex B
(normative)

**Extruded ceramic tiles $3 \% < E_V \leq 6 \%$ Group All_a — Subgroup
(Part) 1**

Dimensional and surface quality requirements and physical and chemical properties are required to be in accordance with Table B.1.

Table B.1 — Requirements for extruded ceramic tiles — Group All_a-1, $3 \% < E_V \leq 6 \%$

1	2	3	4
Dimensions and surface quality	Precision	Natural	Test
Length and width			
The manufacturer shall choose the work size as follows: a) for modular tiles in order to allow a nominal joint width of between 3 mm and 11 mm ^a ; b) for non-modular tiles so that the difference between the work size and the nominal size is not more than ± 3 mm.			
The deviation, in percent, of the average size for each tile (two or four sides) from the work size, S_w	$\pm 1.25 \%$ to a maximum of ± 2 mm	$\pm 2.0 \%$ to a maximum of ± 4 mm	DMS 1701-2
The deviation, in percent, of the average size for each tile (two or four sides) from the average size of the 10 test specimens (20 or 40 sides)	$\pm 1.0 \%$	$\pm 1.5 \%$	DMS 1701-2
Thickness			
a) The thickness shall be specified by the manufacturer ⁹			
b) The deviation, in percent, of the average thickness of each tile from the work size thickness ⁹	$\pm 10 \%$	$\pm 10 \%$	DMS 1701-2
Straightness of sides^b (facial sides)			
The maximum deviation from straightness, in percent, related to the corresponding work sizes	$\pm 0.5 \%$	$\pm 0.6 \%$	DMS 1701-2
Rectangularity^b			
The maximum deviation from rectangularity, in percent, related to the corresponding work sizes	$\pm 1.0 \%$	$\pm 1.0 \%$	DMS 1701-2
Surface flatness			
The maximum deviation from flatness, in percent:			
a) centre curvature, related to diagonal calculated from the work sizes;	$\pm 0.5 \%$	$\pm 1.5 \%$	DMS 1701-2
b) edge curvature, related to the corresponding work	$\pm 0.5 \%$	$\pm 1.5 \%$	DMS 1701-2

Table B.1 (continued)

Dimensions and surface quality	Precision	Natural	Test
c) warpage, related to diagonal calculated from the work sizes.	±0.8 %	±1.5 %	DMS 1701-2
Back feet (if specified)			
a) Height, h , for tiles of surface area, A			
$49 \text{ cm}^2 \leq A < 60 \text{ cm}^2$	Minimum $h = 0.7 \text{ mm}$; Maximum $h = 3.5 \text{ mm}$		Figure 3
$A \geq 60 \text{ cm}^2$	Minimum $h = 1.5 \text{ mm}$; Maximum $h = 3.5 \text{ mm}$		Figure 3
b) Shape	Back feet as specified by the manufacturer and as shown in one of the examples in Figure 3		Figure 3
Example 1 (see Figure 3)	$L_0 - L_1 > 0$		Figure 3
Example 2 (see Figure 3)	$L_0 - L_2 > 0$		Figure 3
Example 3 (see Figure 3)	$L_0 - L_3 > 0$		Figure 3
Surface quality^c	A minimum of 95 % of the tiles are to be free from visible defects which can impair the appearance of a major area of tiles		DMS 1701-2
Physical property	Precision	Natural	Test
Water absorption Percent mass fraction	$3.0 \% < E_v \leq 6.0 \%$ Individual maximum 6.5 %	$3.0 \% < E_v \leq 6.0 \%$ Individual maximum 6.5 %	DMS 1701-3
Breaking strength, in Newtons			
a) Thickness $\geq 7.5 \text{ mm}$	Not less than 950	Not less than 950	DMS 1701-4
b) Thickness $< 7.5 \text{ mm}$	Not less than 600	Not less than 600	DMS 1701-4
Modulus of rupture, in Newtons per square millimetre Not applicable to tiles with breaking strength $\geq 3\,000 \text{ N}$	Minimum 20 Individual minimum 18	Minimum 20 Individual minimum 18	DMS 1701-4
Abrasion resistance			
a) Resistance to deep abrasion of unglazed tiles: removed volume, in cubic millimetres	Maximum 393	Maximum 393	DMS 1701-6
b) Resistance to surface abrasion of glazed tiles intended for use on floors ^d	Report abrasion class and cycles passed	Report abrasion class and cycles passed	DMS 1701-7
Coefficient of linear thermal expansion^e			
From ambient temperature to $100 \text{ }^\circ\text{C}$	Test method available	Test method available	DMS 1701-8
Thermal shock resistance^e	Test method available	Test method available	DMS 1701-9
Crazing resistance: glazed tiles^f	Required	Required	DMS 1701-11
Frost resistance^e	Test method available	Test method available	DMS 1701-12
Moisture expansion, in millimetres per metre^e	Test method available	Test method available	DMS 1701-10
Small colour differences^e	Plain coloured tiles only where required GL: $\Delta E < 0.75$ UGL: $\Delta E < 1.0$	Plain coloured tiles only where required GL: $\Delta E < 0.75$ UGL: $\Delta E < 1.0$	DMS 1701-16

Impact resistance^e	Test method available	Test method available	DMS 1701-5
Chemical property	Precision	Natural	Test
Resistance to staining			
a) Glazed tiles	Minimum class 3	Minimum class 3	DMS 1701-14
b) Unglazed tiles ^e	Test method available	Test method available	DMS 1701-14
Dimensions and surface quality	Precision	Natural	Test
Resistance to chemicals			
Resistance to low concentrations of acids and alkalis	Manufacturer is to state classification	Manufacturer is to state classification	DMS 1701-13
Resistance to high concentrations of acids and alkalis ^e	Test method available	Test method available	DMS 1701-13
Resistance to household chemicals and swimming pool salts	Minimum B	Minimum B	DMS 1701-13
Lead and cadmium release^e	Test method available	Test method available	DMS 1701-15
a Similar joint widths are to be used to apply to traditional systems based on non-metric sizes. b Not applicable to tiles having curved shapes. c As a result of firing, slight variations from this standard colour are unavoidable. This does not apply to intentional irregularities of colour variation of the face of tiles (which can be unglazed, glazed or partly glazed) or to the colour variation over a tile area which is characteristic for this type of tile and desirable. Spots or coloured dots which are introduced for decorative purposes are not considered a defect. d Use Annex N for the abrasion resistance classification for all glazed tiles intended for use on floors. e Annex P gives information regarding requirements which are non-compulsory but which are listed as “test method available”. f Certain decorative effects can have a tendency to craze. These are to be identified by the manufacturer, in which case the crazing test given in DMS 1701-11 is not applicable. g Where applicable, tile thickness shall include the addition of back feet heights, as measured in accordance with Figure 3.			

Annex C
(normative)

**Extruded ceramic tiles $3 \% < E_V \leq 6 \%$ Group All_a — Subgroup
(Part) 2**

Dimensional and surface quality requirements and physical and chemical properties are required to be in accordance with Table C.1.

Table C.1 — Requirements for extruded ceramic tiles — Group All_a-2, $3 \% < E_V \leq 6 \%$

1	2	3	4
Dimensions and surface quality	Precision	Natural	Test
Length and width			
The manufacturer shall choose the work size as follows: a) for modular tiles in order to allow a nominal joint width of between 3 mm and 11 mm ^a ; b) for non-modular tiles so that the difference between the work size and the nominal size is not more than ± 3 mm.			
The deviation, in percent, of the average size for each tile (two or four sides) from the work size, S_W	$\pm 1.5 \%$ to a maximum of ± 2 mm	$\pm 2.0 \%$ to a maximum of ± 4 mm	DMS 1701-2
The deviation, in percent, of the average size for each tile (two or four sides) from the average size of the 10 test specimens (20 or 40 sides)	$\pm 1.5 \%$	$\pm 1.5 \%$	DMS 1701-2
Thickness			
a) The thickness shall be specified by the manufacturer ⁹			
b) The deviation, in percent, of the average thickness of each tile from the work size thickness ⁹	$\pm 10 \%$	$\pm 10 \%$	DMS 1701-2
Straightness of sides^b (facial sides)			
The maximum deviation from straightness, in percent, related to the corresponding work sizes	$\pm 1.0 \%$	$\pm 1.0 \%$	DMS 1701-2
Rectangularity^b			
The maximum deviation from rectangularity, in percent related to the corresponding work sizes	$\pm 1.0 \%$	$\pm 1.0 \%$	DMS 1701-2
Surface flatness			
The maximum deviation from flatness, in percent:			
a) centre curvature, related to diagonal calculated from the work sizes;	$\pm 1.0 \%$	$\pm 1.5 \%$	DMS 1701-2
b) edge curvature, related to the corresponding work sizes;	$\pm 1.0 \%$	$\pm 1.5 \%$	DMS 1701-2

Table C.1 (continued)

Dimensions and surface quality	Precision	Natural	Test
c) warpage, related to diagonal calculated from the work sizes.	±1.5 %	±1.5 %	DMS 1701-2
Back feet (if specified)			
a) Height, h , for tiles of surface area, A			
$49 \text{ cm}^2 \leq A < 60 \text{ cm}^2$	Minimum $h = 0.7 \text{ mm}$; Maximum $h = 3.5 \text{ mm}$		Figure 3
$A \geq 60 \text{ cm}^2$	Minimum $h = 1.5 \text{ mm}$; Maximum $h = 3.5 \text{ mm}$		Figure 3
b) Shape	Back feet as specified by the manufacturer and as shown in one of the examples in Figure 3		Figure 3
Example 1 (see Figure 3)	$L_0 - L_1 > 0$		Figure 3
Example 2 (see Figure 3)	$L_0 - L_2 > 0$		Figure 3
Example 3 (see Figure 3)	$L_0 - L_3 > 0$		Figure 3
Surface quality^c	A minimum of 95 % of the tiles are to be free from visible defects which can impair the appearance of a major area of tiles		DMS 1701-2
Physical property	Precision	Natural	Test
Water absorption Percent mass fraction	$3.0 \% < E_v \leq 6.0 \%$ Individual maximum 6.5 %	$3.0 \% < E_v \leq 6.0 \%$ Individual maximum 6.5 %	DMS 1701-3
Breaking strength, in Newtons			
a) Thickness $\geq 7.5 \text{ mm}$	Not less than 800	Not less than 800	DMS 1701-4
b) Thickness $< 7.5 \text{ mm}$	Not less than 600	Not less than 600	DMS 1701-4
Modulus of rupture, in Newtons per square millimetre Not applicable to tiles with breaking strength $\geq 3\,000 \text{ N}$	Minimum 13 Individual minimum 11	Minimum 13 Individual minimum 11	DMS 1701-4
Abrasion resistance			
a) Resistance to deep abrasion of unglazed tiles: removed volume, in cubic millimetres	Maximum 541	Maximum 541	DMS 1701-6
b) Resistance to surface abrasion of glazed tiles intended for use on floors ^d	Report abrasion class and cycles passed	Report abrasion class and cycles passed	DMS 1701-7
Coefficient of linear thermal expansion^e			
From ambient temperature to $100 \text{ }^\circ\text{C}$	Test method available	Test method available	DMS 1701-8
Thermal shock resistance^e	Test method available	Test method available	DMS 1701-9
Crazing resistance: glazed tiles ^f	Required	Required	DMS 1701-11
Frost resistance^e	Test method available	Test method available	DMS 1701-12
Moisture expansion, in millimetres per metre^e	Test method available	Test method available	DMS 1701-10
Small colour differences^e	Plain coloured tiles only where required GL: $\Delta E < 0.75$ UGL: $\Delta E < 1.0$	Plain coloured tiles only where required GL: $\Delta E < 0.75$ UGL: $\Delta E < 1.0$	DMS 1701-16

Impact resistance^e	Test method available	Test method available	DMS 1701-5
Chemical property	Precision	Natural	Test
Resistance to staining			
a) Glazed tiles	Minimum class 3	Minimum class 3	DMS 1701-14
b) Unglazed tiles ^e	Test method available	Test method available	DMS 1701-14
Dimensions and surface quality	Precision	Natural	Test
Resistance to chemicals			
Resistance to low concentrations of acids and alkalis	Manufacturer is to state classification	Manufacturer is to state classification	DMS 1701-13
Resistance to high concentrations of acids and alkalis ^e	Test method available	Test method available	DMS 1701-13
Resistance to household chemicals and swimming pool salts	Minimum B	Minimum B	DMS 1701-13
Lead and cadmium release^e	Test method available	Test method available	DMS 1701-15
a Similar joint widths are to be used to apply to traditional systems based on non-metric sizes. b Not applicable to tiles having curved shapes. c As a result of firing, slight variations from this standard colour are unavoidable. This does not apply to intentional irregularities of colour variation of the face of tiles (which can be unglazed, glazed or partly glazed) or to the colour variation over a tile area which is characteristic for this type of tile and desirable. Spots or coloured dots which are introduced for decorative purposes are not considered a defect. d Use Annex N for the abrasion resistance classification for all glazed tiles intended for use on floors. e Annex P gives information regarding requirements which are non-compulsory but which are listed as “test method available”. f Certain decorative effects can have a tendency to craze. These are to be identified by the manufacturer, in which case the crazing test given in DMS 1701-11 is not applicable. g Where applicable, tile thickness shall include the addition of back feet heights, as measured in accordance with Figure 3.			

Annex D
(normative)

**Extruded ceramic tiles $6 \% < E_v \leq 10 \%$ Group All_b — Subgroup
(Part) 1**

Dimensional and surface quality requirements and physical and chemical properties are required to be in accordance with Table D.1.

Table D.1 — Requirements for extruded ceramic tiles — Group All_b-1, $6 \% < E_v \leq 10 \%$

1	2	3	4
Dimensions and surface quality	Precision	Natural	Test
Length and width			
The manufacturer shall choose the work size as follows: a) for modular tiles in order to allow a nominal joint width of between 3 mm and 11 mm ^a ; b) for non-modular tiles so that the difference between the work size and the nominal size is not more than ± 3 mm.			
The deviation, in percent, of the average size for each tile (two or four sides) from the work size, S_w	$\pm 2.0 \%$ to a maximum of ± 2 mm	$\pm 2.0 \%$ to a maximum of ± 4 mm	DMS 1701-2
The deviation, in percent, of the average size for each tile (two or four sides) from the average size of the 10 test specimens (20 or 40 sides)	$\pm 1.5 \%$	$\pm 1.5 \%$	DMS 1701-2
Thickness			
a) The thickness shall be specified by the manufacturer ⁹			
b) The deviation, in percent, of the average thickness of each tile from the work size thickness ⁹	$\pm 10 \%$	$\pm 10 \%$	DMS 1701-2
Straightness of sides^b (facial sides)			
The maximum deviation from straightness, in percent, related to the corresponding work sizes	$\pm 1.0 \%$	$\pm 1.0 \%$	DMS 1701-2
Rectangularity^b			
The maximum deviation from rectangularity, in percent, related to the corresponding work sizes	$\pm 1.0 \%$	$\pm 1.0 \%$	DMS 1701-2
Surface flatness			
The maximum deviation from flatness, in percent:			
a) centre curvature, related to diagonal calculated from the work sizes;	$\pm 1.0 \%$	$\pm 1.5 \%$	DMS 1701-2
b) edge curvature, related to the corresponding work	$\pm 1.0 \%$	$\pm 1.5 \%$	DMS 1701-2

Table D.1 (continued)

Dimensions and surface quality	Precision	Natural	Test
c) warpage, related to diagonal calculated from the work sizes.	±1.5 %	±1.5 %	DMS 1701-2
Back feet (if specified)			
a) Height, h , for tiles of surface area, A			
$49 \text{ cm}^2 \leq A < 60 \text{ cm}^2$	Minimum $h = 0.7 \text{ mm}$; Maximum $h = 3.5 \text{ mm}$		Figure 3
$A \geq 60 \text{ cm}^2$	Minimum $h = 1.5 \text{ mm}$; Maximum $h = 3.5 \text{ mm}$		Figure 3
b) Shape	Back feet as specified by the manufacturer and as shown in one of the examples in Figure 3		Figure 3
Example 1 (see Figure 3)	$L_0 - L_1 > 0$		Figure 3
Example 2 (see Figure 3)	$L_0 - L_2 > 0$		Figure 3
Example 3 (see Figure 3)	$L_0 - L_3 > 0$		Figure 3
Surface quality^c	A minimum of 95 % of the tiles are to be free from visible defects which can impair the appearance of a major area of tiles		DMS 1701-2
Physical property	Precision	Natural	Test
Water absorption Percent mass fraction	$6 \% < E_v \leq 10 \%$ Individual maximum 11 %	$6 \% < E_v \leq 10 \%$ Individual maximum 11 %	DMS 1701-3
Breaking strength, in Newtons	Not less than 900	Not less than 900	DMS 1701-4
Modulus of rupture, in Newtons per square millimetre Not applicable to tiles with breaking strength $\geq 3\,000 \text{ N}$	Minimum 17.5 Individual minimum 15	Minimum 17.5 Individual minimum 15	DMS 1701-4
Abrasion resistance			
a) Resistance to deep abrasion of unglazed tiles: removed volume, in cubic millimetres	Maximum 649	Maximum 649	DMS 1701-6
b) Resistance to surface abrasion of glazed tiles intended for use on floors ^d	Report abrasion class and cycles passed	Report abrasion class and cycles passed	DMS 1701-7
Coefficient of linear thermal expansion^e			
From ambient temperature to $100 \text{ }^\circ\text{C}$	Test method available	Test method available	DMS 1701-8
Thermal shock resistance^e	Test method available	Test method available	DMS 1701-9
Crazing resistance: glazed tiles^f	Required	Required	DMS 1701-11
Frost resistance^e	Test method available	Test method available	DMS 1701-12
Moisture expansion, in millimetres per metre	Test method available	Test method available	DMS 1701-10
Small colour differences^e	Plain coloured tiles only where required GL: $\Delta E < 0.75$ UGL: $\Delta E < 1.0$	Plain coloured tiles only where required GL: $\Delta E < 0.75$ UGL: $\Delta E < 1.0$	DMS 1701-16
Impact resistance^e	Test method available	Test method available	DMS 1701-5

Chemical property	Precision	Natural	Test
Resistance to staining			
a) Glazed tiles	Minimum class 3	Minimum class 3	DMS 1701-14
b) Unglazed tiles ^e	Test method available	Test method available	DMS 1701-14
Resistance to chemicals			
Dimensions and surface quality	Precision	Natural	Test
Resistance to low concentrations of acids and alkalis	Manufacturer is to state classification	Manufacturer is to state classification	DMS 1701-13
Resistance to high concentrations of acids and alkalis ^e	Test method available	Test method available	DMS 1701-13
Resistance to household chemicals and swimming pool salts	Minimum B	Minimum B	DMS 1701-13
Lead and cadmium release^e	Test method available	Test method available	DMS 1701-15
a Similar joint widths are to be used to apply to traditional systems based on non-metric sizes. b Not applicable to tiles having curved shapes. c As a result of firing, slight variations from this standard colour are unavoidable. This does not apply to intentional irregularities of colour variation of the face of tiles (which can be unglazed, glazed or partly glazed) or to the colour variation over a tile area which is characteristic for this type of tile and desirable. Spots or coloured dots which are introduced for decorative purposes are not considered a defect. d Use Annex N for the abrasion resistance classification for all glazed tiles intended for use on floors. e Annex P gives information regarding requirements which are non-compulsory but which are listed as "test method available". f Certain decorative effects can have a tendency to craze. These are to be identified by the manufacturer, in which case the crazing test given in DMS 1701-11 is not applicable. g Where applicable, tile thickness shall include the addition of back feet heights, as measured in accordance with Figure 3.			

Annex E
(normative)

**Extruded ceramic tiles $6 \% < E_V \leq 10 \%$ Group All_b — Subgroup
(Part) 2**

Dimensional and surface quality requirements and physical and chemical properties are required to be in accordance with Table E.1.

Table E.1 — Requirements for extruded ceramic tiles — Group All_b-2, $6 \% < E_V \leq 10 \%$

1	2	3	4
Dimensions and surface quality	Precision	Natural	Test
Length and width			
The manufacturer shall choose the work size as follows:			
a) for modular tiles in order to allow a nominal joint width of between 3 mm and 11 mm ^a ;			
b) or non-modular tiles so that the difference between the work size and the nominal size is not more than ± 3 mm.			
The deviation, in percent, of the average size for each tile (two or four sides) from the work size, S_w	$\pm 2.0 \%$ to a maximum of ± 2 mm	$\pm 2.0 \%$ to a maximum of ± 4 mm	DMS 1701-2
The deviation, in percent, of the average size for each tile (two or four sides) from the average size of the 10 test specimens (20 or 40 sides)	$\pm 1.5 \%$	$\pm 1.5 \%$	DMS 1701-2
Thickness			
a) The thickness shall be specified by the manufacturer ⁹			
b) The deviation, in percent, of the average thickness of each tile from the work size thickness ⁹	$\pm 10 \%$	$\pm 10 \%$	DMS 1701-2
Straightness of sides^b (facial sides)			
The maximum deviation from straightness, in percent, related to the corresponding work sizes	$\pm 1.0 \%$	$\pm 1.0 \%$	DMS 1701-2
Rectangularity^b			
The maximum deviation from rectangularity, in percent, related to the corresponding work sizes	$\pm 1.0 \%$	$\pm 1.0 \%$	DMS 1701-2
Surface flatness			
The maximum deviation from flatness, in percent:			
a) centre curvature, related to diagonal calculated from the work sizes;	$\pm 1.0 \%$	$\pm 1.5 \%$	DMS 1701-2
b) edge curvature, related to the corresponding work sizes;	$\pm 1.0 \%$	$\pm 1.5 \%$	DMS 1701-2

Table E.1 (continued)

Dimensions and surface quality	Precision	Natural	Test
c) warpage, related to diagonal calculated from the work sizes.	$\pm 1.5 \%$	$\pm 1.5 \%$	DMS 1701-2
Back feet (if specified)			
a) Height, h , for tiles of surface area, A			
$49 \text{ cm}^2 \leq A < 60 \text{ cm}^2$	Minimum $h = 0.7 \text{ mm}$; Maximum $h = 3.5 \text{ mm}$		Figure 3
$A \geq 60 \text{ cm}^2$	Minimum $h = 1.5 \text{ mm}$; Maximum $h = 3.5 \text{ mm}$		Figure 3
b) Shape	Back feet as specified by the manufacturer and as shown in one of the examples in Figure 3		Figure 3
Example 1 (see Figure 3)	$L_0 - L_1 > 0$		Figure 3
Example 2 (see Figure 3)	$L_0 - L_2 > 0$		Figure 3
Example 3 (see Figure 3)	$L_0 - L_3 > 0$		Figure 3
Surface quality^c	A minimum of 95 % of the tiles are to be free from visible defects which can impair the appearance of a major area of tiles		DMS 1701-2
Physical property	Precision	Natural	Test
Water absorption Percent mass fraction	$6 \% < E_v \leq 10 \%$ Individual maximum 11 %	$6 \% < E_v \leq 10 \%$ Individual maximum 11 %	DMS 1701-3
Breaking strength, in Newtons	Not less than 750	Not less than 750	DMS 1701-4
Modulus of rupture, in Newtons per square millimetre Not applicable to tiles with breaking strength $\geq 3\,000 \text{ N}$.	Minimum 9 Individual minimum 8	Minimum 9 Individual minimum 8	DMS 1701-4
Abrasion resistance			
a) Resistance to deep abrasion of unglazed tiles: removed volume, in cubic millimetres	Maximum 1 062	Maximum 1 062	DMS 1701-6
b) Resistance to surface abrasion of glazed tiles intended for use on floors ^d	Report abrasion class and cycles passed	Report abrasion class and cycles passed	DMS 1701-7
Coefficient of linear thermal expansion^e			
From ambient temperature to $100 \text{ }^\circ\text{C}$	Test method available	Test method available	DMS 1701-8
Thermal shock resistance^e	Test method available	Test method available	DMS 1701-9
Crazing resistance: glazed tiles^f	Required	Required	DMS 1701-11
Frost resistance^e	Test method available	Test method available	DMS 1701-12
Moisture expansion, in millimetres per metre	Test method available	Test method available	DMS 1701-10
Small colour differences^e	Plain coloured tiles only where required GL: $\Delta E < 0.75$ UGL: $\Delta E < 1.0$	Plain coloured tiles only where required GL: $\Delta E < 0.75$ UGL: $\Delta E < 1.0$	DMS 1701-16

Table E.1 (continued)

Dimensions and surface quality	Precision	Natural	Test
Impact resistance^e	Test method available	Test method available	DMS 1701-5
Chemical property	Precision	Natural	Test
Resistance to staining			
a) Glazed tiles	Minimum class 3	Minimum class 3	DMS 1701-14
b) Unglazed tiles ^e	Test method available	Test method available	DMS 1701-14
Resistance to chemicals			
Resistance to low concentrations of acids and alkalis	Manufacturer is to state classification	Manufacturer is to state classification	DMS 1701-13
Resistance to high concentrations of acids and alkalis ^e	Test method available	Test method available	DMS 1701-13
Resistance to household chemicals and swimming pool salts	Minimum B	Minimum B	DMS 1701-13
Lead and cadmium release^e	Test method available	Test method available	DMS 1701-15
a Similar joint widths are to be used to apply to traditional systems based on non-metric sizes. b Not applicable to tiles having curved shapes. c As a result of firing, slight variations from this standard colour are unavoidable. This does not apply to intentional irregularities of colour variation of the face of tiles (which can be unglazed, glazed or partly glazed) or to the colour variation over a tile area which is characteristic for this type of tile and desirable. Spots or coloured dots which are introduced for decorative purposes are not considered a defect. d Use Annex N for the abrasion resistance classification for all glazed tiles intended for use on floors. e Annex P gives information regarding requirements which are non-compulsory but which are listed as "test method available". f Certain decorative effects can have a tendency to craze. These are to be identified by the manufacturer, in which case the crazing test given in DMS 1701-11 is not applicable. g Where applicable, tile thickness shall include the addition of back feet heights, as measured in accordance with Figure 3.			

Annex F
(normative)

Extruded ceramic tiles $E_V > 10\%$ Group AIII

Dimensional and surface quality requirements and physical and chemical properties are required to be in accordance with Table F.1.

Table F.1 — Requirements for extruded ceramic tiles — Group AIII, $E_V > 10\%$

1	2	3	4
Dimensions and surface quality	Precision	Natural	Test
Length and width			
The manufacturer shall choose the work size as follows:			
a) for modular tiles in order to allow a nominal joint width of between 3 mm and 11 mm ^a ;			
b) for non-modular tiles so that the difference between the work size and the nominal size is not more than ± 3 mm.			
The deviation, in percent, of the average size for each tile (two or four sides) from the work size, S_W	$\pm 2.0\%$ to a maximum of ± 2 mm	$\pm 2.0\%$ to a maximum of ± 4 mm	DMS 1701-2
The deviation, in percent, of the average size for each tile (two or four sides) from the average size of the 10 test specimens (20 or 40 sides)	$\pm 1.5\%$	$\pm 1.5\%$	DMS 1701-2
Thickness			
a) The thickness shall be specified by the manufacturer ⁹			
b) The deviation, in percent, of the average thickness of each tile from the work size thickness ⁹	$\pm 10\%$	$\pm 10\%$	DMS 1701-2
Straightness of sides^b (facial sides)			
The maximum deviation from straightness, in percent, related to the corresponding work sizes	$\pm 1.0\%$	$\pm 1.0\%$	DMS 1701-2
Rectangularity^b			
The maximum deviation from rectangularity, in percent, related to the corresponding work sizes	$\pm 1.0\%$	$\pm 1.0\%$	DMS 1701-2
Surface flatness			
The maximum deviation from flatness, in percent:			
a) centre curvature, related to diagonal calculated from the work sizes;	$\pm 1.0\%$	$\pm 1.5\%$	DMS 1701-2
b) edge curvature, related to the corresponding work sizes;	$\pm 1.0\%$	$\pm 1.5\%$	DMS 1701-2

Table F.1 (continued)

Dimensions and surface quality	Precision	Natural	Test
c) warpage, related to diagonal calculated from the work sizes.	±1.5 %	±1.5 %	DMS 1701-2
Back feet (if specified)			
a) Height, h , for tiles of surface area, A			
$49 \text{ cm}^2 \leq A < 60 \text{ cm}^2$	Minimum $h = 0.7 \text{ mm}$; Maximum $h = 3.5 \text{ mm}$		Figure 3
$A \geq 60 \text{ cm}^2$	Minimum $h = 1.5 \text{ mm}$; Maximum $h = 3.5 \text{ mm}$		Figure 3
b) Shape	Back feet as specified by the manufacturer and as shown in one of the examples in Figure 3		Figure 3
Example 1 (see Figure 3)	$L_0 - L_1 > 0$		Figure 3
Example 2 (see Figure 3)	$L_0 - L_2 > 0$		Figure 3
Example 3 (see Figure 3)	$L_0 - L_3 > 0$		Figure 3
Surface quality^c	A minimum of 95 % of the tiles are to be free from visible defects which can impair the appearance of a major area of tiles		DMS 1701-2
Physical property	Precision	Natural	Test
Water absorption Percent mass fraction	$E_v > 10 \%$	$E_v > 10 \%$	DMS 1701-3
Breaking strength, in Newtons	Not less than 600	Not less than 600	DMS 1701-4
Modulus of rupture, in Newtons per square millimetre Not applicable to tiles with breaking strength $\geq 3\,000 \text{ N}$	Minimum 8 Individual minimum 7	Minimum 8 Individual minimum 7	DMS 1701-4
Abrasion resistance			
a) Resistance to deep abrasion of unglazed tiles: removed volume, in cubic millimetres	Maximum 2 365	Maximum 2 365	DMS 1701-6
b) Resistance to surface abrasion of glazed tiles intended for use on floors ^d	Report abrasion class and cycles passed	Report abrasion class and cycles passed	DMS 1701-7
Coefficient of linear thermal expansion^e			
From ambient temperature to $100 \text{ }^\circ\text{C}$	Test method available	Test method available	DMS 1701-8
Thermal shock resistance^e	Test method available	Test method available	DMS 1701-9
Crazing resistance: glazed tiles ^f	Required	Required	DMS 1701-11
Frost resistance^e	Test method available	Test method available	DMS 1701-12
Moisture expansion, in millimetres per metre^e	Test method available	Test method available	DMS 1701-10
Small colour differences^e	Plain coloured tiles only where required GL: $\Delta E < 0.75$ UGL: $\Delta E < 1.0$	Plain coloured tiles only where required GL: $\Delta E < 0.75$ UGL: $\Delta E < 1.0$	DMS 1701-16

Table F.1 (continued)

Dimensions and surface quality	Precision	Natural	Test
Impact resistance^e	Test method available	Test method available	DMS 1701-5
Chemical property	Precision	Natural	Test
Resistance to staining			
a) Glazed tiles	Minimum class 3	Minimum class 3	DMS 1701-14
b) Unglazed tiles ^e	Test method available	Test method available	DMS 1701-14
Resistance to chemicals			
Resistance to low concentrations of acids and alkalis	Manufacturer is to state classification	Manufacturer is to state classification	DMS 1701-13
Resistance to high concentrations of acids and alkalis ^e	Test method available	Test method available	DMS 1701-13
Resistance to household chemicals and swimming pool salts	Minimum B	Minimum B	DMS 1701-13
Lead and cadmium release^e	Test method available	Test method available	DMS 1701-15
a Similar joint widths are to be used to apply to traditional systems based on non-metric sizes. b Not applicable to tiles having curved shapes. c As a result of firing, slight variations from this standard colour are unavoidable. This does not apply to intentional irregularities of colour variation of the face of tiles (which can be unglazed, glazed or partly glazed) or to the colour variation over a tile area which is characteristic for this type of tile and desirable. Spots or coloured dots which are introduced for decorative purposes are not considered a defect. d Use Annex N for the abrasion resistance classification for all glazed tiles intended for use on floors. e Annex P gives information regarding requirements which are non-compulsory but which are listed as “test method available”. f Certain decorative effects can have a tendency to craze. These are to be identified by the manufacturer, in which case the crazing test given in DMS 1701-11 is not applicable. g Where applicable, tile thickness shall include the addition of back feet heights, as measured in accordance with Figure 3.			

Annex G
(normative)

Dry-pressed ceramic tiles with low water absorption $E_v \leq 0.5\%$ Group BI_a

Dimensional and surface quality requirements and physical and chemical properties are required to be in accordance with Table G.1.

Table G.1 — Requirements for dry-pressed ceramic tiles with low water absorption — Group BI_a, $E_v \leq 0.5\%$

Dimensions and surface quality	Nominal size N			Test
	7 cm ≤ N < 15 cm	N ≥ 15 cm		
	mm	%	mm	
Length and width		whichever is less is applicable		
The manufacturer shall choose the work size as follows: for modular tiles in order to allow a nominal joint width of between 2 mm and 5 mm ^a ; for non-modular tiles so that the difference between the work size and the nominal size is not more than ±2 % (max. ±5 mm).				
For non-rectified, the deviation of the average size for each tile (two or four sides) from the work size, S _w	±0.9	±0.6	±2.0	DMS 1701-2
For rectified tiles, the deviation of the average size for each tile (two or four sides) from the work size, S _w	±0.4	±0.3	±1.0	DMS 1701-2
Thickness				
The thickness shall be specified by the manufacturer ^h				
The deviation of the average thickness of each tile from the work size thickness ^h	±0.5	±5	±0.5	DMS 1701-2
Straightness of sides ^b (facial sides)				
For non-rectified tiles, the maximum deviation from straightness related to the corresponding work sizes	±0.8	±0.5	±1.5	DMS 1701-2
For rectified tiles, the maximum deviation from straightness related to the corresponding work sizes	±0.4	±0.3	± 0.8	DMS 1701-2
Rectangularity ^{b, i}				
For non-rectified tiles, the maximum deviation from rectangularity related to the corresponding work sizes	±0.8	±0.5	±2.0	DMS 1701-2

Table G.1 (continued)

Dimensions and surface quality	Nominal size N			Test
	7 cm ≤ N < 15 cm	N ≥ 15 cm		
	mm	%	mm	
For rectified tiles, the maximum deviation from rectangularity related to the corresponding work sizes	±0.4	±0.3	±1.5	DMS 1701-2
Surface flatness		whichever is less is applicable		
The maximum deviation from flatness:				
a) for non-rectified tiles, centre curvature, related to diagonal calculated from the work sizes;	±0.8	±0.5	±2.0	DMS 1701-2
b) for rectified tiles, centre curvature, related to diagonal calculated from the work sizes;	±0.6	±0.4	±1.8	DMS 1701-2
c) for non-rectified tiles, edge curvature, related to the corresponding work sizes;	±0.8	±0.5	±2.0	DMS 1701-2
d) for rectified tiles, edge curvature, related to the corresponding work sizes;	±0.6	±0.4	±1.8	DMS 1701-2
e) for non-rectified tiles, warpage, related to diagonal calculated from the work sizes;	±0.8	±0.5	±2.0	DMS 1701-2
f) for rectified tiles, warpage, related to diagonal calculated from the work sizes.	±0.6	±0.4	±1.8	DMS 1701-2
Back feet (if specified)				
a) Height, <i>h</i> , for tiles of surface area, <i>A</i>				
49 cm ² ≤ <i>A</i> < 60 cm ²	Minimum <i>h</i> = 0.7 mm; Maximum <i>h</i> = 3.5 mm			Figure 3
<i>A</i> ≥ 60 cm ²	Minimum <i>h</i> = 1.5 mm; Maximum <i>h</i> = 3.5 mm			Figure 3
b) Shape	Back feet as specified by the manufacturer and as shown in one of the examples in Figure 3			Figure 3
Example 1 (see Figure 3)	<i>L</i> ₀ − <i>L</i> ₁ > 0			Figure 3
Example 2 (see Figure 3)	<i>L</i> ₀ − <i>L</i> ₂ > 0			Figure 3
Example 3 (see Figure 3)	<i>L</i> ₀ − <i>L</i> ₃ > 0			Figure 3
Surface quality ^c	A minimum of 95 % of the tiles are to be free from visible defects which can impair the appearance of a major area of tiles			DMS 1701-2
Physical property	Requirements			Test
Water absorption Percent mass fraction ^g	<i>E</i> _v ≤ 0.5 % Individual maximum 0.6 %			DMS 1701-3
Breaking strength, in Newtons				
a) Thickness ≥ 7.5 mm	Not less than 1 300			DMS 1701-4
b) Thickness < 7.5 mm	Not less than 700			DMS 1701-4

Table G.1 (continued)

Dimensions and surface quality	Nominal size N			Test
	7 cm ≤ N < 15 cm	N ≥ 15 cm		
	mm	%	mm	
Modulus of rupture, in Newtons per square millimetre Not applicable to tiles with breaking strength ≥3 000 N	Minimum 35 Individual minimum 32			DMS 1701-4
Abrasion resistance				
a) Resistance to deep abrasion of unglazed tiles: removed volume, in cubic millimetres	Maximum 175			DMS 1701-6
b) Resistance to surface abrasion of glazed tiles intended for use on floors ^d	Report abrasion class and cycles passed			DMS 1701-7
Coefficient of linear thermal expansion^e				
From ambient temperature to 100 °C	Test method available			DMS 1701-8
Thermal shock resistance^e	Test method available			DMS 1701-9
Crazing resistance: glazed tiles ^f	Required			DMS 1701-11
Frost resistance	Required			DMS 1701-12
Moisture expansion, in millimetres per metre^e	Test method available			DMS 1701-10
Small colour differences^e	Plain coloured tiles only where required GL: ΔE < 0.75 UGL: ΔE < 1.0			DMS 1701-16
Impact resistance^e	Test method available			DMS 1701-5
Chemical property	Requirements			Test
Resistance to staining				
a) Glazed tiles	Minimum class 3			DMS 1701-14
b) Unglazed tiles ^e	Test method available			DMS 1701-14
Resistance to chemicals				
Resistance to low concentrations of acids and alkalis	Manufacturer is to state classification			DMS 1701-13
Resistance to high concentrations of acids and alkalis ^e	Test method available			DMS 1701-13

Table G.1 (continued)

Dimensions and surface quality	Nominal size N			Test
	7 cm ≤ N < 15 cm	N ≥ 15 cm		
	mm	%	mm	
Resistance to household chemicals and swimming pool salts	Minimum B			DMS 1701-13
Lead and cadmium release ^e	Test method available			DMS 1701-15
a Similar joint widths are to be used to apply to traditional systems based on non-metric sizes. b Not applicable to tiles having curved shapes. c As a result of firing, slight variations from this standard colour are unavoidable. This does not apply to intentional irregularities of colour variation of the face of tiles (which can be unglazed, glazed or partly glazed) or to the colour variation over a tile area which is characteristic for this type of tile and desirable. Spots or coloured dots which are introduced for decorative purposes are not considered a defect. d Use Annex N for the abrasion resistance classification for all glazed tiles intended for use on floors. e Annex P gives information regarding requirements which are non-compulsory but which are listed as “test method available”. f Certain decorative effects can have a tendency to craze. These are to be identified by the manufacturer, in which case the crazing test given in DMS 1701-11 is not applicable. g A fully vitrified tile is a tile with water absorption of a maximum individual value of 0.5 % (porcelain tile). h Where applicable, tile thickness shall include the addition of back feet heights, as measured in accordance with Figure 3 i For oblong tiles with longer edge ≥60 cm, and ratio between longer edge and shorter edge ≥3, only d_L and percent deviation d_L/S are applicable. See DMS 1701-2				

Annex H
(normative)

Dry-pressed ceramic tiles with low water absorption $0.5 \% < E_V \leq 3 \%$ Group BIb

Dimensional and surface quality requirements and physical and chemical properties are required to be in accordance with Table H.1.

Table H.1 — Requirements for dry-pressed ceramic tiles with low water absorption — Group BIb, $0.5 \% < E_V \leq 3 \%$

1	2	3	4	5
Dimensions and surface quality	Nominal size N			Test
	7 cm ≤ N < 15 cm	N ≥ 15 cm		
	mm	%	mm	
Length and width		whichever is less is applicable		
The manufacturer shall choose the work size as follows: a) for modular tiles in order to allow a nominal joint width of between 2 mm and 5 mm ^a ; b) for non-modular tiles so that the difference between the work size and the nominal size is not more than ±2 % (max. ±5 mm).				
For non-rectified tiles, the deviation of the average size for each tile (two or four sides) from the work size, S _w	±0.9	±0.6	±2.0	DMS 1701-2
For rectified tiles, the deviation of the average size for each tile (two or four sides) from the work size, S _w	±0.4	±0.3	±1.0	DMS 1701-2
Thickness				
a) The thickness shall be specified by the manufacturer ^g				
b) The deviation of the average thickness of each tile from the work size thickness ^g	±0.5	±5	±0.5	DMS 1701-2
Straightness of sides ^b (facial sides)				
For non-rectified tiles, the maximum deviation from straightness related to the corresponding work sizes	±0.8	±0.5	±1.5	DMS 1701-2
For rectified tiles, the maximum deviation from straightness related to the corresponding work sizes	±0.4	±0.3	± 0.8	DMS 1701-2
Rectangularity ^{b, h}				
For non-rectified tiles, the maximum deviation from rectangularity related to the corresponding work sizes	±0.8	±0.5	±2.0	DMS 1701-2

Table H.1 (continued)

Dimensions and surface quality	Nominal size N			Test
	7 cm ≤ N < 15 cm	N ≥ 15 cm		
	mm	%	mm	
For rectified tiles, the maximum deviation from rectangularity related to the corresponding work sizes	±0.4	±0.3	±1.5	DMS 1701-2
Surface flatness		whichever is less is applicable		
The maximum deviation from flatness:				
a) for non-rectified tiles, centre curvature, related to diagonal calculated from the work sizes;	±0.8	±0.5	±2.0	DMS 1701-2
b) for rectified tiles, centre curvature, related to diagonal calculated from the work sizes;	±0.6	±0.4	±1.8	DMS 1701-2
c) for non-rectified tiles, edge curvature, related to the corresponding work sizes;	±0.8	±0.5	±2.0	DMS 1701-2
d) for rectified tiles, edge curvature, related to the corresponding work sizes;	±0.6	±0.4	±1.8	DMS 1701-2
e) for non-rectified tiles, warpage, related to diagonal calculated from the work sizes;	±0.8	±0.5	±2.0	DMS 1701-2
f) for rectified tiles, warpage, related to diagonal calculated from the work sizes.	±0.6	±0.4	±1.8	DMS 1701-2
Back feet (if specified)				
a) Height, <i>h</i> , for tiles of surface area, <i>A</i>				
49 cm ² ≤ <i>A</i> < 60 cm ²	Minimum <i>h</i> = 0.7 mm; Maximum <i>h</i> = 3.5 mm			Figure 3
<i>A</i> ≥ 60 cm ²	Minimum <i>h</i> = 1.5 mm; Maximum <i>h</i> = 3.5 mm			Figure 3
b) Shape	ack feet as specified by the manufacturer and as shown in one of the examples in Figure 3			Figure 3
Example 1 (see Figure 3)	<i>L</i> ₀ − <i>L</i> ₁ > 0			Figure 3
Example 2 (see Figure 3)	<i>L</i> ₀ − <i>L</i> ₂ > 0			Figure 3
Example 3 (see Figure 3)	<i>L</i> ₀ − <i>L</i> ₃ > 0			Figure 3
Surface quality ^c	A minimum of 95 % of the tiles are to be free from visible defects which can impair the appearance of a major area of tiles			DMS 1701-2
Physical property	Requirements			Test
Water absorption Percent mass fraction	0.5 % < <i>E</i> _V ≤ 3 % Individual maximum 3.3 %			DMS 1701-3
Breaking strength, in Newtons				
a) Thickness ≥7.5 mm	Not less than 1 100			DMS 1701-4
b) Thickness <7.5 mm	Not less than 700			DMS 1701-4
Modulus of rupture, in Newtons per square millimetre Not applicable to tiles with breaking strength ≥3 000 N	Minimum 30 Individual minimum 27			DMS 1701-4
Abrasion resistance				
a) Resistance to deep abrasion of unglazed tiles: removed volume, in cubic millimetres	Maximum 175			DMS 1701-6

Table H.1 (continued)

Dimensions and surface quality	Nominal size N			Test
	7 cm ≤ N < 15 cm	N ≥ 15 cm		
	mm	%	mm	
b) Resistance to surface abrasion of glazed tiles intended for use on floors ^d	Report abrasion class and cycles passed			DMS 1701-7
Coefficient of linear thermal expansion^e				
From ambient temperature to 100 °C	Test method available			DMS 1701-8
Thermal shock resistance^e	Test method available			DMS 1701-9
Crazing resistance: glazed tiles ^f	Required			DMS 1701-11
Frost resistance	Required			DMS 1701-12
Moisture expansion, in millimetres per metre^e	Test method available			DMS 1701-10
Small colour differences^e	Plain coloured tiles only where required GL: $\Delta E < 0.75$ UGL: $\Delta E < 1.0$			DMS 1701-16
Impact resistance^e	Test method available			DMS 1701-5
Chemical property	Requirements			Test
Resistance to staining				
a) Glazed tiles	Minimum class 3			DMS 1701-14
b) Unglazed tiles ^e	Test method available			DMS 1701-14
Resistance to chemicals				
Resistance to low concentrations of acids and alkalis	Manufacturer is to state classification			DMS 1701-13
Resistance to high concentrations of acids and alkalis ^e	Test method available			DMS 1701-13
Resistance to household chemicals and swimming pool salts	Minimum B			DMS 1701-13
Lead and cadmium release^e	Test method available			DMS 1701-15
a Similar joint widths are to be used to apply to traditional systems based on non-metric sizes.				
b Not applicable to tiles having curved shapes.				
c As a result of firing, slight variations from this standard colour are unavoidable. This does not apply to intentional irregularities of colour variation of the face of tiles (which can be unglazed, glazed or partly glazed) or to the colour variation over a tile area which is characteristic for this type of tile and desirable. Spots or coloured dots which are introduced for decorative purposes are not considered a defect.				
d Use Annex N for the abrasion resistance classification for all glazed tiles intended for use on floors.				
e Annex P gives information regarding requirements which are non-compulsory but which are listed as “test method available”.				
f Certain decorative effects can have a tendency to craze. These are to be identified by the manufacturer, in which case the crazing test given in DMS 1701-11 is not applicable.				
g Where applicable, tile thickness shall include the addition of back feet heights, as measured in accordance with Figure 3.				
h For rectangular tiles with longer edge ≥60 cm, and ratio between longer edge and shorter edge ≥3, only d_L and percent deviation d_L/S are applicable. See DMS 1701-2				

Annex I
(informative)

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This annex is intentionally left blank. This is as a convenience to producers in order to avoid changes to packaging and related costs. In addition, at the time of publication, the market offers consumers various products identified by the matching annex headings of this document.

DRAFT MALAWI STANDARD

Annex J
(normative)

Dry-pressed ceramic tiles $3 \% < E_v \leq 6 \%$ Group BII_a

Dimensional and surface quality requirements and physical and chemical properties are required to be in accordance with Table J.1

Table J.1— Requirements for dry-pressed ceramic tiles — Group BII_a, $3 \% < E_v \leq 6 \%$

1	2	3	4	5
Dimensions and surface quality	Nominal size N			Test
	7 cm ≤ N < 15 cm	N ≥ 15 cm		
	mm	%	mm	
Length and width		whichever is less is applicable		
The manufacturer shall choose the work size as follows: a) for modular tiles in order to allow a nominal joint width of between 2 mm and 5 mm ^a ; b) for non-modular tiles so that the difference between the work size and the nominal size is not more than ±2 % (max. ±5 mm).				
For non-rectified tiles, the deviation of the average size for each tile (two or four sides) from the work size, S _w	±0.9	±0.6	±2.0	DMS 1701-2
For rectified tiles, the deviation of the average size for each tile (two or four sides) from the work size, S _w	±0.4	±0.3	±1.0	DMS 1701-2
Thickness				
The thickness shall be specified by the manufacturer ⁹				
The deviation of the average thickness of each tile from the work size thickness ⁹	±0.5	±5	±0.5	DMS 1701-2
Straightness of sides ^b (facial sides)				
For non-rectified tiles, the maximum deviation from straightness related to the corresponding work sizes	±0.8	±0.5	±1.5	DMS 1701-2
For rectified tiles, the maximum deviation from straightness related to the corresponding work sizes	±0.4	±0.3	± 0.8	DMS 1701-2
Rectangularity ^{b, h}				
For non-rectified tiles, the maximum deviation from rectangularity related to the corresponding work sizes	±0.8	±0.5	±2.0	DMS 1701-2
For rectified tiles, the maximum deviation from rectangularity related to the corresponding work sizes	±0.4	±0.3	±1.5	DMS 1701-2

Table J.1 (continued)

Dimensions and surface quality	Nominal size N			Test
	7 cm ≤ N < 15 cm	N ≥ 15 cm		
	mm	%	mm	
Surface flatness		whichever is less is applicable		
The maximum deviation from flatness:				
a) for non-rectified tiles, centre curvature, related to diagonal calculated from the work sizes;	±0.8	±0.5	±2.0	DMS 1701-2
b) for rectified tiles, centre curvature, related to diagonal calculated from the work sizes;	±0.6	±0.4	±1.8	DMS 1701-2
c) for non-rectified tiles, edge curvature, related to the corresponding work sizes;	±0.8	±0.5	±2.0	DMS 1701-2
d) for rectified tiles, edge curvature, related to the corresponding work sizes;	±0.6	±0.4	±1.8	DMS 1701-2
e) for non-rectified tiles, warpage, related to diagonal calculated from the work sizes.	±0.8	±0.5	±2.0	DMS 1701-2
f) for rectified tiles, warpage, related to diagonal calculated from the work sizes.	±0.6	±0.4	±1.8	DMS 1701-2
Back feet (if specified)				
a) Height, <i>h</i> , for tiles of surface area, <i>A</i>				
49 cm ² ≤ <i>A</i> < 60 cm ²	Minimum <i>h</i> = 0.7 mm; Maximum <i>h</i> = 3.5 mm			Figure 3
<i>A</i> ≥ 60 cm ²	Minimum <i>h</i> = 1.5 mm; Maximum <i>h</i> = 3.5 mm			Figure 3
b) Shape	Back feet as specified by the manufacturer and as shown in one of the examples in Figure 3			Figure 3
Example 1 (see Figure 3)	<i>L</i> ₀ – <i>L</i> ₁ > 0			Figure 3
Example 2 (see Figure 3)	<i>L</i> ₀ – <i>L</i> ₂ > 0			Figure 3
Example 3 (see Figure 3)	<i>L</i> ₀ – <i>L</i> ₃ > 0			Figure 3
Surface quality ^c	A minimum of 95 % of the tiles are to be free from visible defects which can impair the appearance of a major area of tiles			DMS 1701-2
Physical property	Requirements			Test
Water absorption Percent mass fraction	3 % < <i>E_V</i> ≤ 6 % Individual maximum 6.5 %			DMS 1701-3
Breaking strength, in Newtons				
a) Thickness ≥7.5 mm	Not less than 1 000			DMS 1701-4
b) Thickness <7.5 mm	Not less than 600			DMS 1701-4
Modulus of rupture, in Newtons per square millimetre Not applicable to tiles with breaking strength ≥3 000 N	Minimum 22 Individual minimum 20			DMS 1701-4
Abrasion resistance				
a) Resistance to deep abrasion of unglazed tiles: removed volume, in cubic millimetres	Maximum 345			DMS 1701-6
b) Resistance to surface abrasion of glazed tiles intended for use on floors ^d	Report abrasion class and cycles passed			DMS 1701-7
Coefficient of linear thermal expansion ^e				

Table J.1 (continued)

Dimensions and surface quality	Nominal size N			Test
	7 cm ≤ N < 15 cm	N ≥ 15 cm		
	mm	%	mm	
From ambient temperature to 100 °C	Test method available			DMS 1701-8
Thermal shock resistance ^e	Test method available			DMS 1701-9
Crazing resistance: glazed tiles ^f	Required			DMS 1701-11
Frost resistance ^e	Test method available			DMS 1701-12
Moisture expansion, in millimetres per metre ^e	Test method available			DMS 1701-10
Small colour differences ^e	Plain coloured tiles only where required GL: $\Delta E < 0.75$ UGL: $\Delta E < 1.0$			DMS 1701-16
Impact resistance ^e	Test method available			DMS 1701-5
Chemical property	Requirements			Test
Resistance to staining				
a) Glazed tiles	Minimum class 3			DMS 1701-14
b) Unglazed tiles ^e	Test method available			DMS 1701-14
Resistance to chemicals				
Resistance to low concentrations of acids and alkalis	Manufacturer is to state classification			DMS 1701-13
Resistance to high concentrations of acids and alkalis ^e	Test method available			DMS 1701-13
Resistance to household chemicals and swimming pool salts	Minimum B			DMS 1701-13
Lead and cadmium release ^e	Test method available			DMS 1701-15
a Similar joint widths are to be used to apply to traditional systems based on non-metric sizes.				
b Not applicable to tiles having curved shapes.				
c As a result of firing, slight variations from this standard colour are unavoidable. This does not apply to intentional irregularities of colour variation of the face of tiles (which can be unglazed, glazed or partly glazed) or to the colour variation over a tile area which is characteristic for this type of tile and desirable. Spots or coloured dots which are introduced for decorative purposes are not considered a defect.				
d Use Annex N for the abrasion resistance classification for all glazed tiles intended for use on floors.				
e Annex P gives information regarding requirements which are non-compulsory but which are listed as “test method available”.				
f Certain decorative effects can have a tendency to craze. These are to be identified by the manufacturer, in which case the crazing test given in DMS 1701-11 is not applicable.				
g Where applicable, tile thickness shall include the addition of back feet heights, as measured in accordance with Figure 3.				
h For rectangular tiles with longer edge ≥60 cm, and ratio between longer edge and shorter edge ≥3, only d _L and percent deviation d _L /S are applicable. See DMS 1701-2				

Annex K
(normative)

Dry-pressed ceramic tiles $6\% < E_v \leq 10\%$ Group BIIb

Dimensional and surface quality requirements and physical and chemical properties are required to be in accordance with Table K.1.

Table K.1 — Requirements for dry-pressed ceramic tiles — Group BIIb, $6\% < E_v \leq 10\%$

1	2	3	4	5
Dimensions and surface quality	Nominal size N			Test
	7 cm ≤ N < 15 cm	N ≥ 15 cm		
	mm	%	mm	
Length and width		whichever is less is applicable		
The manufacturer shall choose the work size as follows: a) for modular tiles in order to allow a nominal joint width of between 2 mm and 5 mm ^a ; b) for non-modular tiles so that the difference between the work size and the nominal size is not more than ±2 % (max. ±5 mm).				
For non-rectified tiles, the deviation of the average size for each tile (two or four sides) from the work size, S _w	±0.9	±0.6	±2.0	DMS 1701-2
For rectified tiles, the deviation of the average size for each tile (two or four sides) from the work size, S _w	±0.4	±0.3	±1.0	DMS 1701-2
Thickness				
a) The thickness shall be specified by the manufacturer ^g				
b) The deviation of the average thickness of each tile from the work size thickness ^g	±0.5	±5	±0.5	DMS 1701-2
Straightness of sides ^b (facial sides)				
For non-rectified tiles, the maximum deviation from straightness related to the corresponding work sizes	±0.8	±0.5	±1.5	DMS 1701-2
For rectified tiles, the maximum deviation from straightness related to the corresponding work sizes	±0.4	±0.3	± 0.8	DMS 1701-2
Rectangularity ^{b, h}				
For non-rectified tiles, the maximum deviation from rectangularity related to the corresponding work sizes	±0.8	±0.5	±2.0	DMS 1701-2
For rectified tiles, the maximum deviation from rectangularity related to the corresponding work sizes	±0.4	±0.3	±1.5	DMS 1701-2
Surface flatness		whichever is less is applicable		

Table K.1 (continued)

Dimensions and surface quality	Nominal size N			Test
	7 cm ≤ N < 15 cm	N ≥ 15 cm		
	mm	%	mm	
The maximum deviation from flatness:				
a) for non-rectified tiles, centre curvature, related to diagonal calculated from the work sizes;	±0.8	±0.5	±2.0	DMS 1701-2
b) for rectified tiles, centre curvature, related to diagonal calculated from the work sizes;	±0.6	±0.4	±1.8	DMS 1701-2
c) for non-rectified tiles, edge curvature, related to the corresponding work sizes;	±0.8	±0.5	±2.0	DMS 1701-2
d) for rectified tiles, edge curvature, related to the corresponding work sizes;	±0.6	±0.4	±1.8	DMS 1701-2
e) for non-rectified tiles, warpage, related to diagonal calculated from the work sizes.	±0.8	±0.5	±2.0	DMS 1701-2
f) for rectified tiles, warpage, related to diagonal calculated from the work sizes.	±0.6	±0.4	±1.8	DMS 1701-2
Back feet (if specified)				
a) Height, <i>h</i> , for tiles of surface area, <i>A</i>				
49 cm ² ≤ <i>A</i> < 60 cm ²	Minimum <i>h</i> = 0.7 mm; Maximum <i>h</i> = 3.5 mm			Figure 3
<i>A</i> ≥ 60 cm ²	Minimum <i>h</i> = 1.5 mm; Maximum <i>h</i> = 3.5 mm			Figure 3
b) Shape	Back feet as specified by the manufacturer and as shown in one of the examples in Figure 3			Figure 3
Example 1 (see Figure 3)	<i>L</i> ₀ − <i>L</i> ₁ > 0			Figure 3
Example 2 (see Figure 3)	<i>L</i> ₀ − <i>L</i> ₂ > 0			Figure 3
Example 3 (see Figure 3)	<i>L</i> ₀ − <i>L</i> ₃ > 0			Figure 3
Surface quality ^c	A minimum of 95 % of the tiles are to be free from visible defects which can impair the appearance of a major area of tiles			DMS 1701-2
Physical property	Requirements			Test
Water absorption Percent mass fraction	6 % < <i>E</i> _v ≤ 10 % Individual maximum 11 %			DMS 1701-3
Breaking strength, in Newtons				
a) Thickness ≥7.5 mm	Not less than 800			DMS 1701-4
b) Thickness <7.5 mm	Not less than 500			DMS 1701-4
Modulus of rupture, in Newtons per square millimetre Not applicable to tiles with breaking strength ≥3 000 N	Minimum 18 Individual minimum 16			DMS 1701-4
Abrasion resistance				
a) Resistance to deep abrasion of unglazed tiles: removed volume, in cubic millimetres	Maximum 540			DMS 1701-6
b) Resistance to surface abrasion of glazed tiles intended for use on floors ^d	Report abrasion class and cycles passed			DMS 1701-7
Coefficient of linear thermal expansion ^e				
From ambient temperature to 100 °C	Test method available			DMS 1701-8
Thermal shock resistance ^e	Test method available			DMS 1701-9
Crazing resistance: glazed tiles ^f	Required			DMS 1701-11

Table K.1 (continued)

Dimensions and surface quality	Nominal size N			Test
	7 cm ≤ N < 15 cm	N ≥ 15 cm		
	mm	%	mm	
Frost resistance ^e	Test method available			DMS 1701-12
Moisture expansion, in millimetres per metre ^e	Test method available			DMS 1701-10
Small colour differences ^e	Plain coloured tiles only where required GL: ΔE < 0.75 UGL: ΔE < 1.0			DMS 1701-16
Impact resistance ^e	Test method available			DMS 1701-5
Chemical property	Requirements			Test
Resistance to staining				
a) Glazed tiles	Minimum class 3			DMS 1701-14
b) Unglazed tiles ^e	Test method available			DMS 1701-14
Resistance to chemicals				
Resistance to low concentrations of acids and alkalis	Manufacturer is to state classification			DMS 1701-13
Resistance to high concentrations of acids and alkalis ^e	Test method available			DMS 1701-13
Resistance to household chemicals and swimming pool salts	Minimum B			DMS 1701-13
Lead and cadmium release ^e	Test method available			DMS 1701-15
a Similar joint widths are to be used to apply to traditional systems based on non-metric sizes.				
b Not applicable to tiles having curved shapes.				
c As a result of firing, slight variations from this standard colour are unavoidable. This does not apply to intentional irregularities of colour variation of the face of tiles (which can be unglazed, glazed or partly glazed) or to the colour variation over a tile area which is characteristic for this type of tile and desirable. Spots or coloured dots which are introduced for decorative purposes are not considered a defect.				
d Use Annex N for the abrasion resistance classification for all glazed tiles intended for use on floors.				
e Annex P gives information regarding requirements which are non-compulsory but which are listed as “test method available”.				
f Certain decorative effects can have a tendency to craze. These are to be identified by the manufacturer, in which case the crazing test given in DMS 1701-11 is not applicable.				
g Where applicable, tile thickness shall include the addition of back feet heights, as measured in accordance with Figure 3.				
h For rectangular tiles with longer edge ≥60 cm, and ratio between longer edge and shorter edge ≥3, only d _L and percent deviation d _L /S are applicable. See DMS 1701-2 .				

Annex L
(normative)

Dry-pressed ceramic tiles $E_V > 10\%$ Group BIII

Dimensional and surface quality requirements and physical and chemical properties are required to be in accordance with Table L.1.

Table L.1 — Requirements for dry-pressed ceramic tiles — Group BIII, $E_V > 10\%$

1	2	3	4	5
Dimensions and surface quality	Nominal size N			Test
	7 cm ≤ N < 15 cm	N ≥ 15 cm		
	mm	%	mm	
Length and width		whichever is less is applicable		
The manufacturer shall choose the work size as follows:				
a) for modular tiles in order to allow a nominal joint width of between 1.5 mm and 5 mm ^a ;				
b) for non-modular tiles so that the difference between the work size and the nominal size is not more than ±2 % (max. ±5 mm).				
For non-rectified tiles, the deviation of the average size for each tile (two or four sides) from the work size, Sw	±0.9	±0.6	±2.0	DMS 1701-2
For rectified tiles, the maximum deviation from straightness related to the corresponding work sizes	±0.4	±0.3	± 1.0	DMS 1701-2
Thickness				
a) The thickness shall be specified by the manufacturer ^h				
b) The deviation of the average thickness of each tile from the work size thickness ^h	±0.5	±10	±0.5	DMS 1701-2
Straightness of sides ^b (facial sides)				
For non-rectified tiles, the maximum deviation from straightness related to the corresponding work sizes	±0.8	±0.5	±1.5	DMS 1701-2
For rectified tiles, the maximum deviation from straightness related to the corresponding work sizes	±0.4	±0.3	± 0.8	DMS 1701-2
Rectangularity ^{b, i}				
For non-rectified tiles, the maximum deviation from rectangularity related to the corresponding work sizes	±0.8	±0.5	±2.0	DMS 1701-2
For rectified tiles, the maximum deviation from rectangularity related to the corresponding work sizes	±0.4	±0.3	±1.5	DMS 1701-2

Table L.1 (continued)

Dimensions and surface quality	Nominal size N			Test
	7 cm ≤ N < 15 cm	N ≥ 15 cm		
	mm	%	mm	
Surface flatness		whichever is less is applicable		
The maximum deviation from flatness:				
a) for non-rectified tiles, centre curvature, related to diagonal calculated from the work sizes;	±0.8	±0.5	±2.0	DMS 1701-2
b) for rectified tiles, centre curvature, related to diagonal calculated from the work sizes;	±0.6	±0.4	±1.8	DMS 1701-2
c) for non-rectified tiles, edge curvature, related to the corresponding work sizes;	±0.8	±0.5	±2.0	DMS 1701-2
d) for rectified tiles, edge curvature, related to the corresponding work sizes;	±0.6	±0.4	±1.8	DMS 1701-2
e) for non-rectified tiles, warpage, related to diagonal calculated from the work sizes.	±0.8	±0.5	±2.0	DMS 1701-2
f) for rectified tiles, warpage, related to diagonal calculated from the work sizes.	±0.6	±0.4	±1.8	DMS 1701-2
Back feet (if specified)				
a) Height, <i>h</i> , for tiles of surface area, <i>A</i>				
49 cm ² ≤ <i>A</i> < 60 cm ²	Minimum <i>h</i> = 0.7 mm; Maximum <i>h</i> = 3.5 mm			Figure 3
<i>A</i> ≥ 60 cm ²	Minimum <i>h</i> = 1.5 mm; Maximum <i>h</i> = 3.5 mm			Figure 3
b) Shape	Back feet as specified by the manufacturer and as shown in one of the examples in Figure 3			Figure 3
Example 1 (see Figure 3)	<i>L</i> ₀ – <i>L</i> ₁ > 0			Figure 3
Example 2 (see Figure 3)	<i>L</i> ₀ – <i>L</i> ₂ > 0			Figure 3
Example 3 (see Figure 3)	<i>L</i> ₀ – <i>L</i> ₃ > 0			Figure 3
Surface quality ^c	A minimum of 95 % of the tiles are to be free from visible defects which can impair the appearance of a major area of tiles			DMS 1701-2
Physical property	Requirements			Test
Water absorption Percent mass fraction	<i>E</i> _v > 10 %. Where the average exceeds 20 %, this shall be indicated by the manufacturer Individual minimum value 9 %			DMS 1701-3
Breaking strength, in Newtons ^g				
a) Thickness ≥7.5 mm	Not less than 600			DMS 1701-4
b) Thickness <7.5 mm	Not less than 200			DMS 1701-4
Modulus of rupture, in Newtons per square millimetre Not applicable to tiles with breaking strength ≥3 000 N				DMS 1701-4
a) Thickness ≥7.5 mm	Minimum 12			DMS 1701-4
b) Thickness <7.5 mm	Minimum 15			DMS 1701-4
Abrasion resistance				
Resistance to surface abrasion of glazed tiles intended for use on floors ^d	Report abrasion class and cycles passed			DMS 1701-7
Coefficient of linear thermal expansion ^e				
From ambient temperature to 100°C	Test method available			DMS 1701-8

Table L.1 (continued)

Dimensions and surface quality	Nominal size N			Test
	7 cm ≤ N < 15 cm	N ≥ 15 cm		
	mm	%	mm	
Thermal shock resistance ^e	Test method available			DMS 1701-9
Crazing resistance: glazed tiles ^f	Required			DMS 1701-11
Frost resistance ^e	Test method available			DMS 1701-12
Moisture expansion, in millimetres per metre ^e	Test method available			DMS 1701-10
Small colour differences ^e	Plain coloured tiles only where required GL: ΔE < 0.75 UGL: ΔE < 1.0			DMS 1701-16
Impact resistance ^e	Test method available			DMS 1701-5
Chemical property	Requirements			Test
Resistance to staining				
a) Glazed tiles	Minimum class 3			DMS 1701-14
Resistance to chemicals				
Resistance to low concentrations of acids and alkalis	Test method available			DMS 1701-13
Resistance to high concentrations of acids and alkalis ^e	Test method available			DMS 1701-13
Resistance to household chemicals and swimming pool salts	Minimum B			DMS 1701-13
Lead and cadmium release ^e	Test method available			DMS 1701-15
a Similar joint widths are to be used to apply to traditional systems based on non-metric sizes.				
b Not applicable to tiles having curved shapes.				
c As a result of firing, slight variations from this standard colour are unavoidable. This does not apply to intentional irregularities of colour variation of the face of tiles (which can be unglazed, glazed or partly glazed) or to the colour variation over a tile area which is characteristic for this type of tile and desirable. Spots or coloured dots which are introduced for decorative purposes are not considered a defect.				
d Use Annex N for the abrasion resistance classification for all glazed tiles intended for use on floors.				
e Annex P gives information regarding requirements which are non-compulsory but which are listed as “test method available”.				
f Certain decorative effects can have a tendency to craze. These are to be identified by the manufacturer, in which case the crazing test given in DMS 1701-11 is not applicable.				
g Tiles with breaking strength less than 400 N are intended for use on walls only and the manufacturer is required to specify the intended use.				
h Where applicable, tile thickness shall include the addition of back feet heights, as measured in accordance with Figure 3.				
i For rectangular tiles with longer edge ≥60 cm, and ratio between longer edge and shorter edge ≥3, only d _L and percent deviation d _L /S are applicable. See DMS 1701-2				

Annex M
(normative)

Extruded ceramic tiles with low water absorption $E_V \leq 0.5\%$ Group A_a

Dimensional and surface quality requirements and physical and chemical properties are required to be in accordance with Table M.1.

Table M.1 — Requirements for extruded ceramic tiles — Group A_a, $E_V < 0.5\%$

1	2	3	4
Dimensions and surface quality	Precision	Natural	Test
Length and width			
The manufacturer shall choose the work size as follows: a) for modular tiles in order to allow a nominal joint width of between 3 mm and 11 mm ^a ; b) for non-modular tiles so that the difference between the work size and the nominal size is not more than ± 3 mm.			
The deviation, in percent, of the average size for each tile (two or four sides) from the work size, S_w	$\pm 1.0\%$ to a maximum of ± 2 mm	$\pm 2.0\%$ to a maximum of ± 4 mm	DMS 1701-2
The deviation, in percent, of the average size for each tile (two or four sides) from the average size of the 10 test specimens (20 or 40 sides)	$\pm 1.0\%$	$\pm 1.5\%$	DMS 1701-2
Thickness			
a) The thickness shall be specified by the manufacturer ^h			
b) The deviation, in percent, of the average thickness of each tile from the work size thickness ^h	$\pm 10\%$	$\pm 10\%$	DMS 1701-2
Straightness of sides^b (facial sides)			
The maximum deviation from straightness, in percent, related to the corresponding work sizes	$\pm 0.5\%$	$\pm 0.6\%$	DMS 1701-2
Rectangularity^b			
The maximum deviation from rectangularity, in percent, related to the corresponding work sizes	$\pm 1.0\%$	$\pm 1.0\%$	DMS 1701-2
Surface flatness			
The maximum deviation from flatness, in percent:			
a) centre curvature, related to diagonal calculated from the work sizes;	$\pm 0.5\%$	$\pm 1.5\%$	DMS 1701-2
b) edge curvature, related to the corresponding work	$\pm 0.5\%$	$\pm 1.5\%$	DMS 1701-2

Table M.1 (continued)

Dimensions and surface quality	Precision	Natural	Test
c) warpage, related to diagonal calculated from the work sizes.	±0.8 %	±1.5 %	DMS 1701-2
Back feet (if specified)			
a) Height, h , for tiles of surface area, A			
$49 \text{ cm}^2 \leq A < 60 \text{ cm}^2$	Minimum $h = 0.7 \text{ mm}$; Maximum $h = 3.5 \text{ mm}$		Figure 3
$A \geq 60 \text{ cm}^2$	Minimum $h = 1.5 \text{ mm}$; Maximum $h = 3.5 \text{ mm}$		Figure 3
b) Shape	Back feet as specified by the manufacturer and as shown in one of the examples in Figure 3		Figure 3
Example 1 (see Figure 3)	$L_0 - L_1 > 0$		Figure 3
Example 2 (see Figure 3)	$L_0 - L_2 > 0$		Figure 3
Example 3 (see Figure 3)	$L_0 - L_3 > 0$		Figure 3
Surface quality^c	A minimum of 95 % of the tiles are to be free from visible defects which can impair the appearance of a major area of tiles		DMS 1701-2
Physical property	Precision	Natural	Test
Water absorption Percent mass fraction ^g	$E_v \leq 0.5 \%$ Individual maximum 0.6 %	$E_v \leq 0.5 \%$ Individual maximum 0.6 %	DMS 1701-3
Breaking strength, in Newtons			
a) Thickness $\geq 7.5 \text{ mm}$	Not less than 1 300	Not less than 1 300	DMS 1701-4
b) Thickness $< 7.5 \text{ mm}$	Not less than 600	Not less than 600	DMS 1701-4
Modulus of rupture, in Newtons per square millimetre Not applicable to tiles with breaking strength $\geq 3\,000 \text{ N}$	Minimum 28 Individual minimum 21	Minimum 28 Individual minimum 21	DMS 1701-4
Abrasion resistance			
a) Resistance to deep abrasion of unglazed tiles: removed volume, in cubic millimetres	Maximum 275	Maximum 275	DMS 1701-6
b) Resistance to surface abrasion of glazed tiles intended for use on floors ^d	Report abrasion class and cycles passed	Report abrasion class and cycles passed	DMS 1701-7
Coefficient of linear thermal expansion^e			
From ambient temperature to $100 \text{ }^\circ\text{C}$	Test method available	Test method available	DMS 1701-8
Thermal shock resistance^e	Test method available	Test method available	DMS 1701-9
Crazing resistance: glazed tiles^f	Required	Required	DMS 1701-11
Frost resistance	Required	Required	DMS 1701-12
Moisture expansion, in millimetres per metre^e	Test method available	Test method available	DMS 1701-10
Small colour differences^e	Plain coloured tiles only where required GL: $\Delta E < 0.75$ UGL: $\Delta E < 1.0$	Plain coloured tiles only where required GL: $\Delta E < 0.75$ UGL: $\Delta E < 1.0$	DMS 1701-16

Table M.1 (continued)

Impact resistance	Test method available	Test method available	DMS 1701-5
Chemical property	Precision	Natural	Test
Resistance to staining			
a) Glazed tiles	Minimum class 3	Minimum class 3	DMS 1701-14
b) Unglazed tiles ^e	Test method available	Test method available	DMS 1701-14
Dimensions and surface quality	Precision	Natural	Test
Resistance to chemicals			
Resistance to low concentrations of acids and alkalis	Manufacturer is to state classification	Manufacturer is to state classification	DMS 1701-13
Resistance to high concentrations of acids and alkalis ^e	Test method available	Test method available	DMS 1701-13
Resistance to household chemicals and swimming pool salts	Minimum B	Minimum B	DMS 1701-13
Lead and cadmium release^e	Test method available	Test method available	DMS 1701-15
a Similar joint widths are to be used to apply to traditional systems based on non-metric sizes. b Not applicable to tiles having curved shapes. c As a result of firing, slight variations from this standard colour are unavoidable. This does not apply to intentional irregularities of colour variation of the face of tiles (which can be unglazed, glazed or partly glazed) or to the colour variation over a tile area which is characteristic for this type of tile and desirable. Spots or coloured dots which are introduced for decorative purposes are not considered a defect. d Use Annex N for the abrasion resistance classification for all glazed tiles intended for use on floors. e Annex P gives information regarding requirements which are non-compulsory but which are listed as “test method available”. f Certain decorative effects can have a tendency to craze. These are to be identified by the manufacturer, in which case the crazing test given in DMS 1701-11 is not applicable. g A fully vitrified tile is a tile with water absorption of a maximum individual value of 0.5 % (porcelain tile). h Where applicable, tile thickness shall include the addition of back feet heights, as measured in accordance with Figure 3.			

Annex N
(informative)

Classification of glazed tiles for floors according to their abrasion resistance

This approximate classification is given for guidance only (see **DMS 1701-7**). It is not to be taken to provide accurate product specifications for specific requirements.

- Class 0 Glazed tiles in this class are not recommended for use on floors.
- Class 1 Floor coverings in areas which are walked on, essentially with soft-soled footwear or bare feet without scratching dirt (for example residential bathrooms and bedrooms without direct access from the outside).
- Class 2 Floor coverings in areas which are walked on by soft-soled or normal footwear, with, at the most, occasional small amounts of scratching dirt (for example rooms in the living areas of homes, but with the exception of kitchens, entrances and other rooms which can have a lot of traffic). This does not apply to abnormal footwear, for example hobnailed boots.
- Class 3 Floor coverings in areas which, with normal footwear, are walked on more often with small amounts of scratching dirt (for example residential kitchens, halls, corridors, balconies, loggias and terraces). This does not apply to abnormal footwear, for example hob-nailed boots.
- Class 4 Floor coverings which are walked on by regular traffic with some scratching dirt so that the conditions are more severe than class 3 (for example entrances, commercial kitchens, hotel, exhibition and sale rooms).
- Class 5 Floor coverings which are subjected to severe pedestrian traffic over sustained periods with some scratching dirt, so that the conditions are the most severe for which glazed floor tiles are to be suitable (for example public areas, such as shopping centres, airport concourses, hotel foyers, public walkways and industrial applications).

This classification is valid for the given applications under normal conditions. Consideration should be given to the footwear, type of traffic and cleaning methods expected, and the floors should be adequately protected against scratching dirt at the entrances to buildings by interposing footwear cleaning devices. In extreme cases of very heavy traffic and quantities of scratching dirt, unglazed floor tiles and quarries from Group I may be considered.

Annex O
(informative)

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This annex is intentionally left blank. This is as a convenience to producers in order to avoid changes to packaging and related costs. In addition, at the time of publication, the market offers consumers various products identified by the matching annex headings of this document.

DRAFT MALAWI STANDARD

Annex P
(informative)

Test methods

A number of test methods that are included in this International Standard are available on request, but do not form part of the compulsory test requirements. The purpose of this annex is to provide explanatory comments on the inclusion of these tests and other pertinent information.

DMS 1701-5: This test is intended only for testing tiles which are used in areas where impact resistance is considered to be of particular importance. The normal requirement for light-duty installations is a coefficient of restitution of 0.55. For heavier duty applications, a higher figure is required.

DMS 1701-8: Most ceramic tiles have low levels of linear thermal expansion. This test is intended for tiles which are installed in conditions of high thermal variation.

DMS 1701-9: All ceramic tiles withstand high temperatures. This test is to be applied to any ceramic tile which is subjected to localized thermal shock.

DMS 1701-10: The majority of glazed and unglazed tiles have negligible moisture expansion, which does not contribute to tiling problems if tiles are correctly fixed (installed). However, with unsatisfactory fixing practices or in certain climatic conditions, moisture expansion in excess of 0.06 % (0.6 mm/m) can contribute to problems.

DMS 1701-12: This test is compulsory only for products which are intended to be specified for use where frost can apply. The test is not required for product groups which are generally unsuitable for use where frost is present.

DMS 1701-13: Ceramic tiles are normally resistant to common chemicals. The test for high concentrations of acids and alkalis is intended for ceramic tiles which are to be used in potentially corrosive conditions.

DMS 1701-14: This test is compulsory for glazed tiles. For unglazed tiles, where staining is a problem, it is recommended that the manufacturer be consulted. This method does not address the temporary colour changes which can occur in certain types of glazed tile due to the absorption of water in the body under the glaze.

DMS 1701-15: This test is intended for glazed tiles which are used on worktops and on wall surfaces where food preparation takes place, and in cases where food is to be in direct contact with the glazed tile surface. For indicative limits, see **Directive 2005/31/EC**.

DMS 1701-16: This test is only applicable to plain coloured glazed or unglazed tiles and is considered to be of importance in certain specialized circumstances. It is to be used only where small colour differences between plain coloured tiles are important in a specification.

Annex Q
(informative)

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Annex R
(informative)
Symbols recommended for use

The use of symbols on packaging and/or literature is not a requirement except where stated; however, the symbols shown in Table R.1 are recommended to indicate intended use.

Table R.1 — Recommended symbols

1	2	3
Symbol	Description/application	ISO/IEC registration number
	ceramic tile suitable for use on floors	ISO_7000_3622
	ceramic tile suitable for use on walls	ISO_7000_3623
	glazed tile intended for use on floors according to its abrasion resistance (see Annex N)	Not applicable
	frost-resistant ceramic tile	ISO_7000_3624

Bibliography

- [1] European Commission, European Commission Directive 2005/31/EC of 29 April 2005 amending Council Directive 84/500/EEC as regards a declaration of compliance and performance criteria of the analytical method for ceramic articles intended to come into contact with foodstuffs
- [2] ISO 7000, *Graphical symbols for use on equipment — Registered symbols*

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